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(54) DISPLAY DEVICE

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(57) **ABSTRACT**

A display device includes a pixel unit including pixels. Each pixel includes a light emitting element, a driving transistor controlling an amount of current flowing from a first power source line to a second power source line via the light emitting element in response to a voltage of a node, a write transistor connected between a first electrode of the driving transistor and a data line and turned on or off in response to a write scan signal, and an initialization transistor connected between an anode electrode of the light emitting element and an initialization power source line and turned on or off in response to an initialization scan signal. A low voltage of the write scan signal is set to a first gate low voltage, and a low voltage of the initialization scan signal is set to a second gate low voltage different from the first gate low voltage.

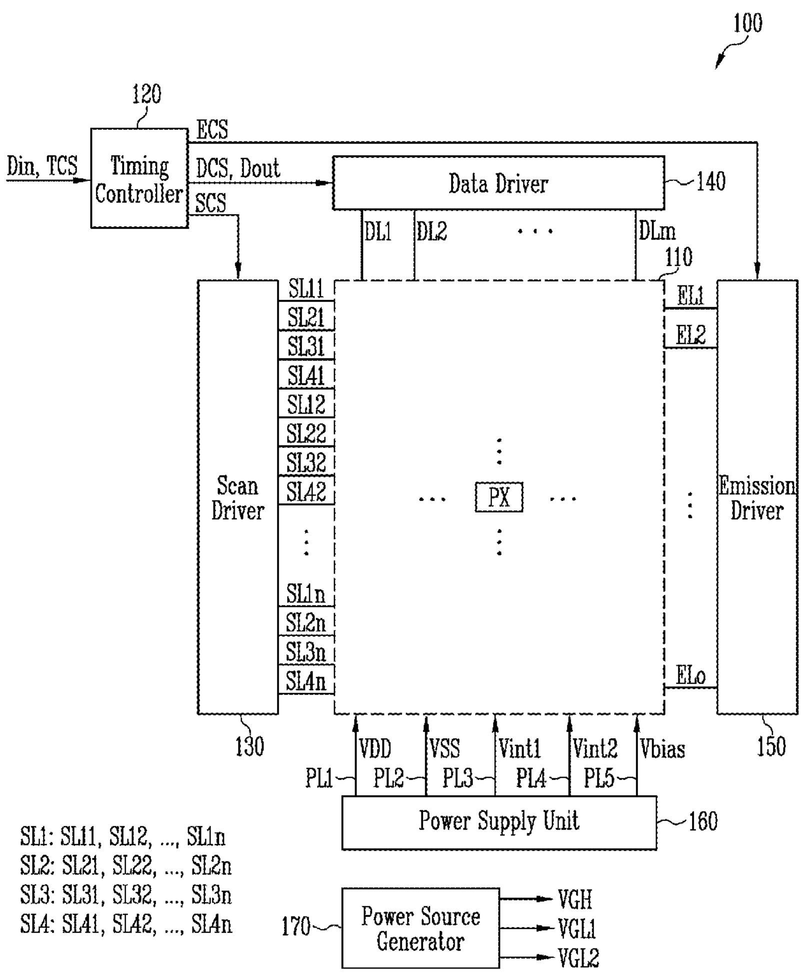
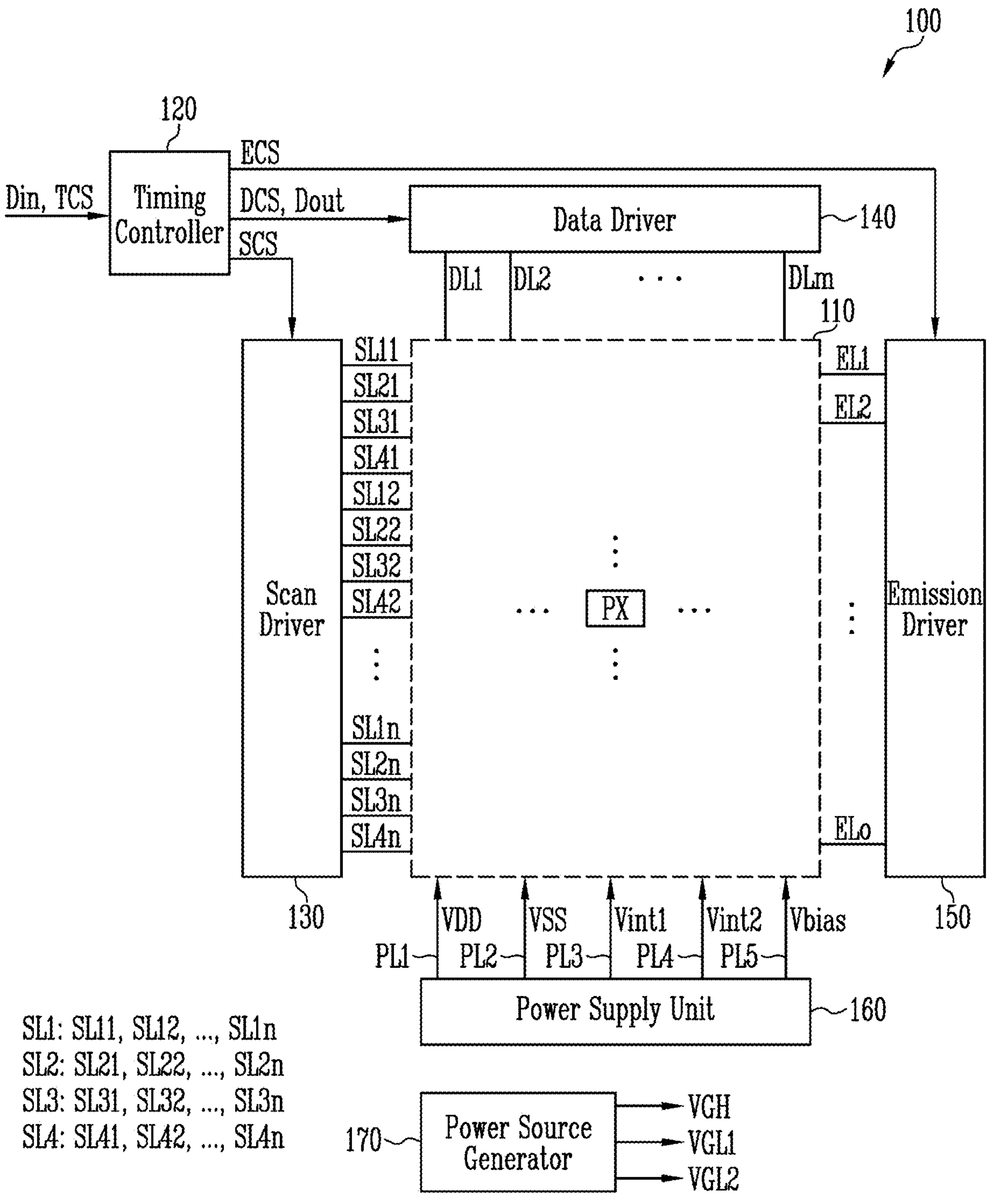


FIG. 1



100
120
Din, TCS
Timing Controller
ECS
DCS, Dout
SCS
Data Driver
140
DL1
DL2
DLm
110
SL11
SL21
SL31
SL41
SL12
SL22
SL32
SL42
SL1n
SL2n
SL3n
SL4n
Scan Driver
PX
EL1
EL2
ELo
Emission Driver
130
150
VDD
VSS
Vint1
Vint2
Vbias
PL1
PL2
PL3
PL4
PL5
Power Supply Unit
160
SL1: SL11, SL12, ..., SL1n
SL2: SL21, SL22, ..., SL2n
SL3: SL31, SL32, ..., SL3n
SL4: SL41, SL42, ..., SL4n
170
Power Source Generator
VGH
VGL1
VGL2

FIG. 2

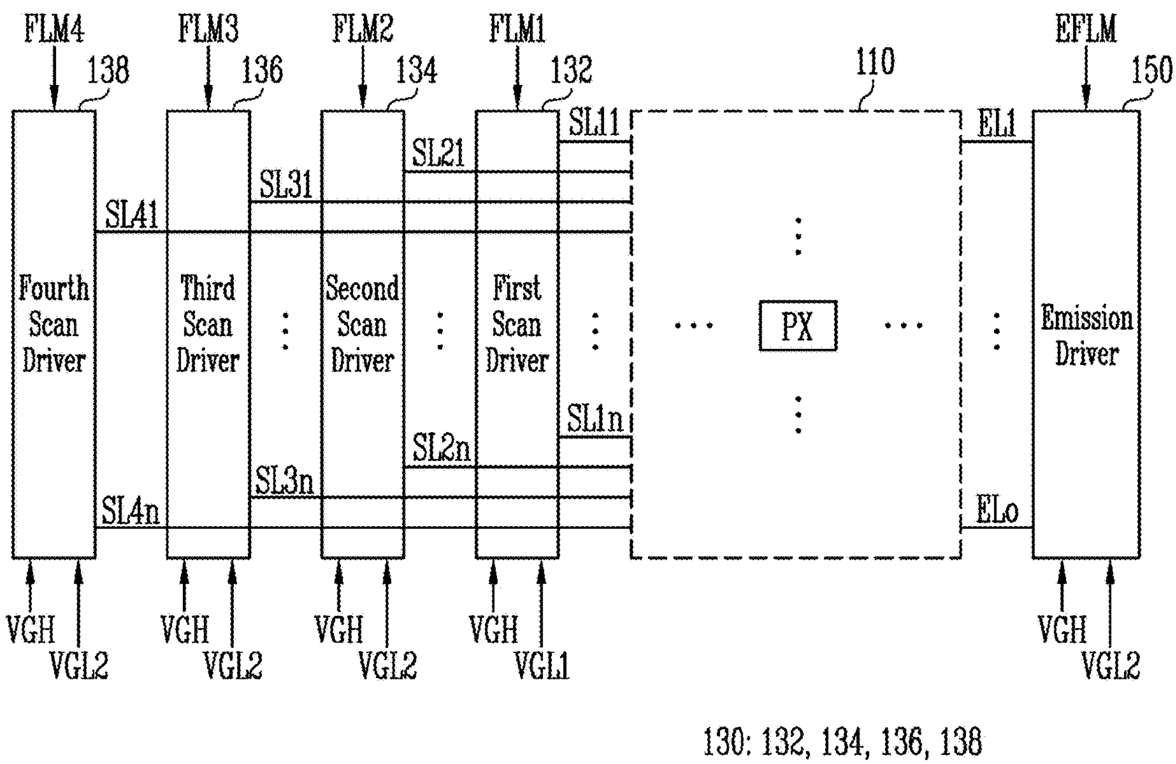
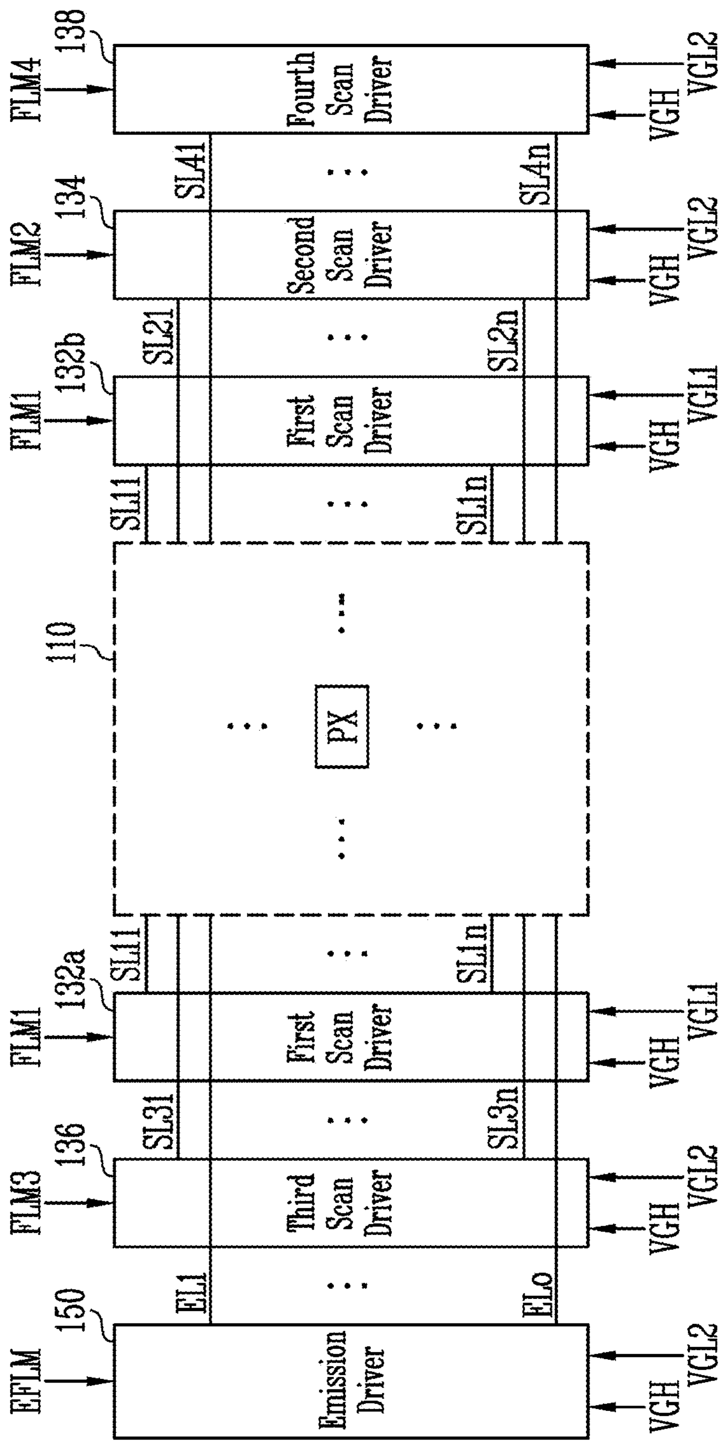

FLM4
138
FLM3
136
FLM2
134
FLM1
132
110
EFLM
150
SL11
SL21
SL31
SL41
EL1
Fourth Scan Driver
Third Scan Driver
Second Scan Driver
First Scan Driver
PX
Emission Driver
SL1n
SL2n
SL3n
SL4n
ELo
VGH
VGL2
VGH
VGL2
VGH
VGL2
VGH
VGL1
VGH
VGL2
130: 132, 134, 136, 138

FIG. 3



130: 132, 134, 136, 138
132: 132a, 132b

FIG. 4

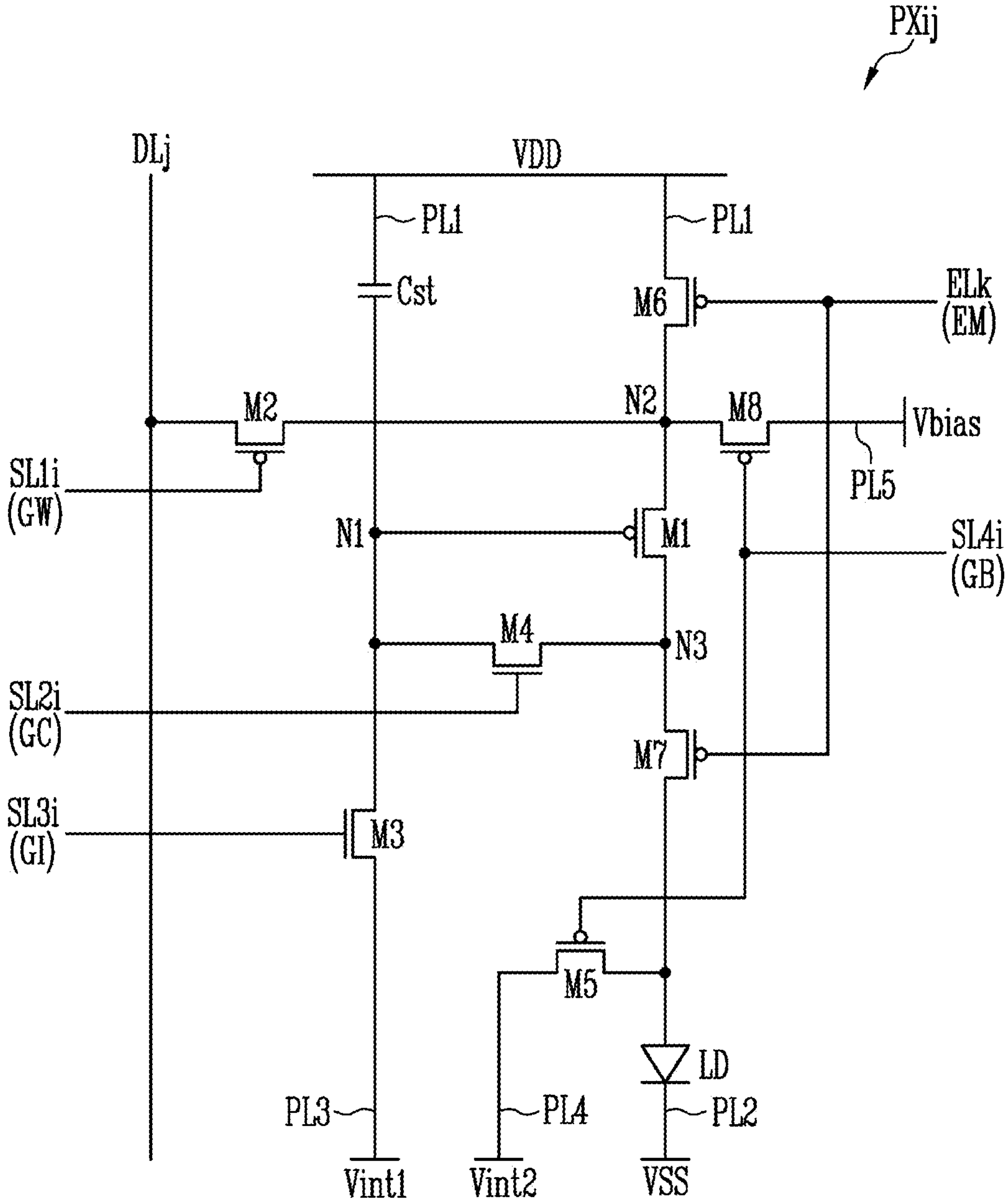

PXij
DLj
VDD
PL1
PL1
Cst
M6
ELk
(EM)
M2
N2
M8
Vbias
SL1i
(GW)
PL5
N1
M1
SL4i
(GB)
M4
N3
SL2i
(GC)
M7
SL3i
(GI)
M3
M5
LD
PL3
PL4
PL2
Vint1
Vint2
VSS

FIG. 5

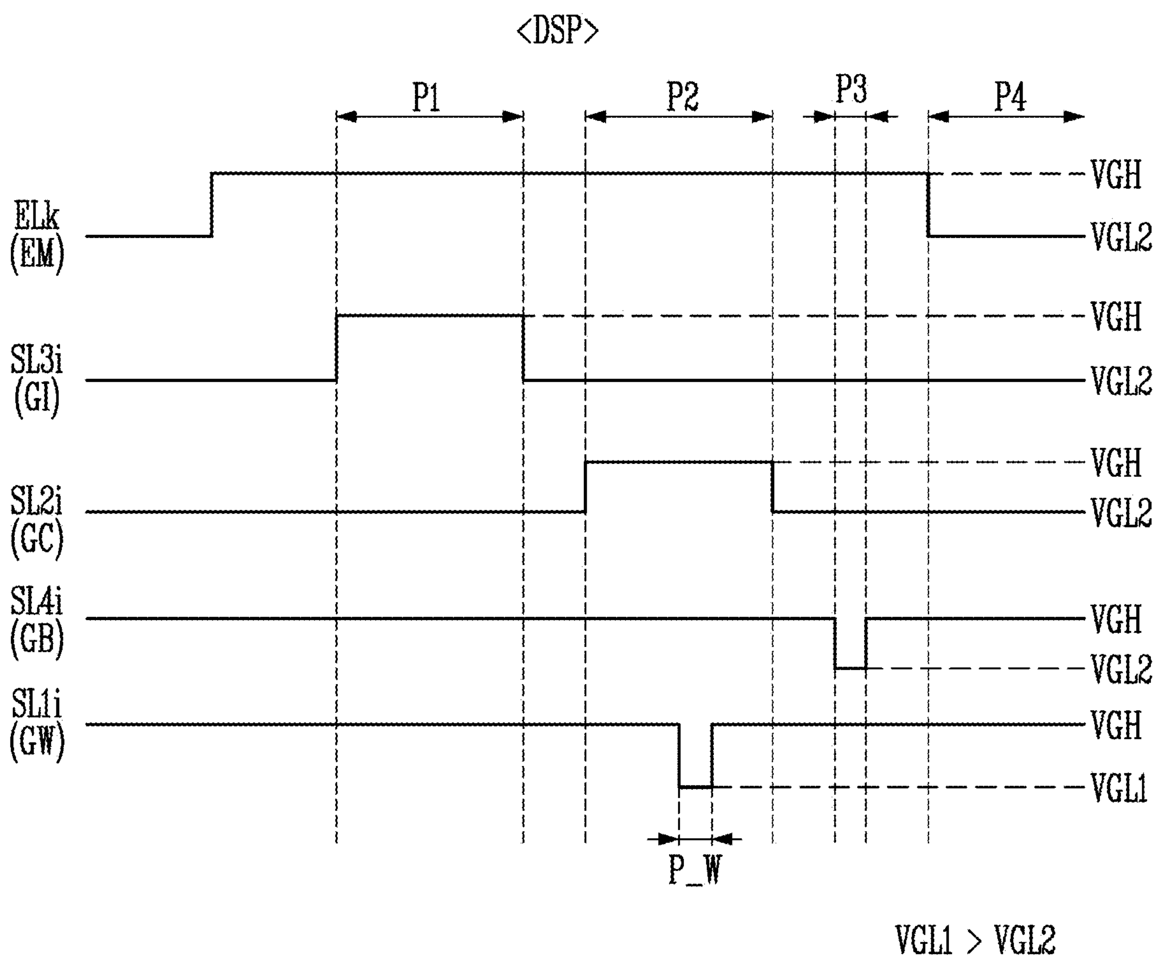

<DSP>
P1
P2
P3
P4
ELk
(EM)
SL3i
(GI)
SL2i
(GC)
SL4i
(GB)
SL1i
(GW)
VGH
VGL2
VGH
VGL2
VGH
VGL2
VGH
VGL2
VGH
VGL1
P_W
VGL1 > VGL2

FIG. 6

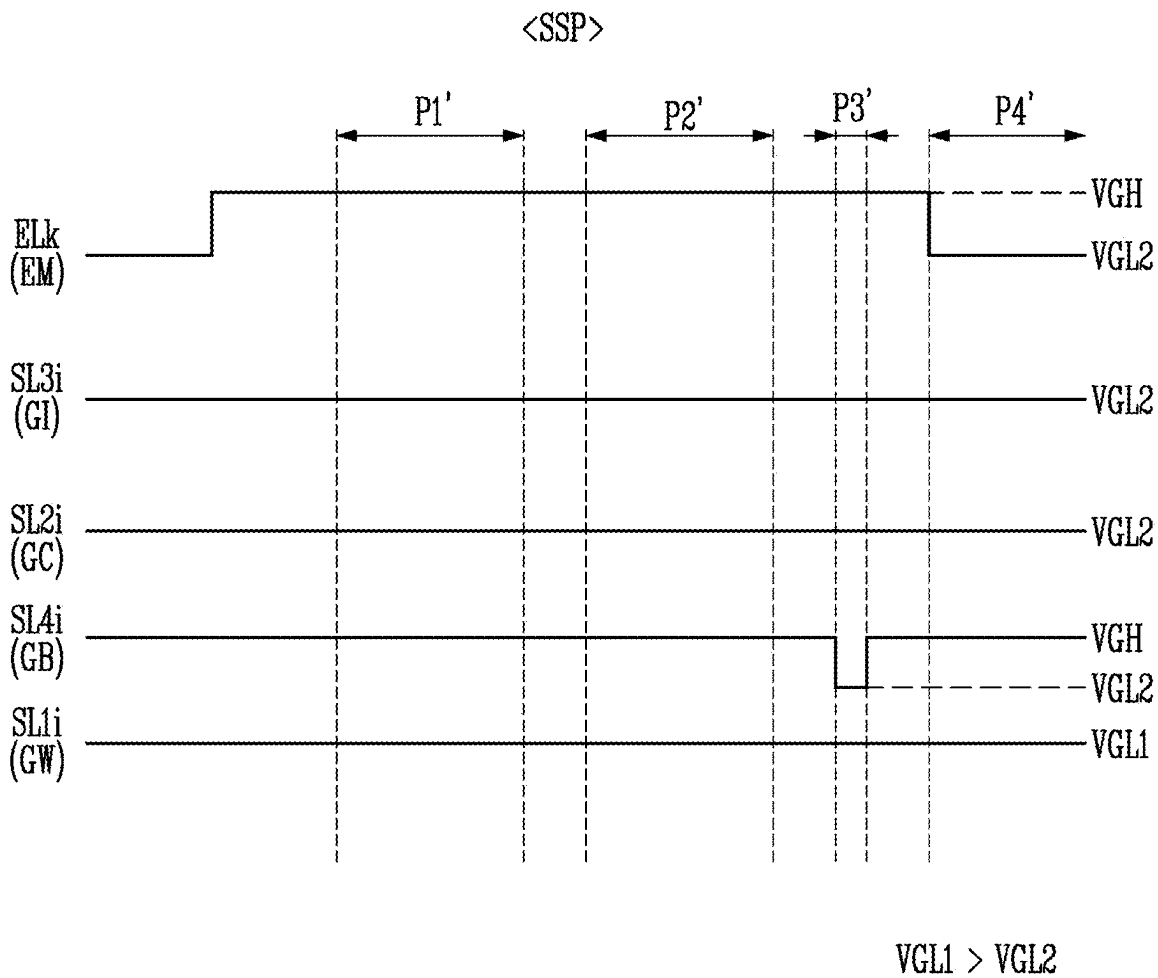

<SSP>
P1'
P2'
P3'
P4'
ELk
(EM)
VGH
VGL2
SL3i
(GI)
VGL2
SL2i
(GC)
VGL2
SL4i
(GB)
VGH
VGL2
SL1i
(GW)
VGL1
VGL1 > VGL2

FIG. 7

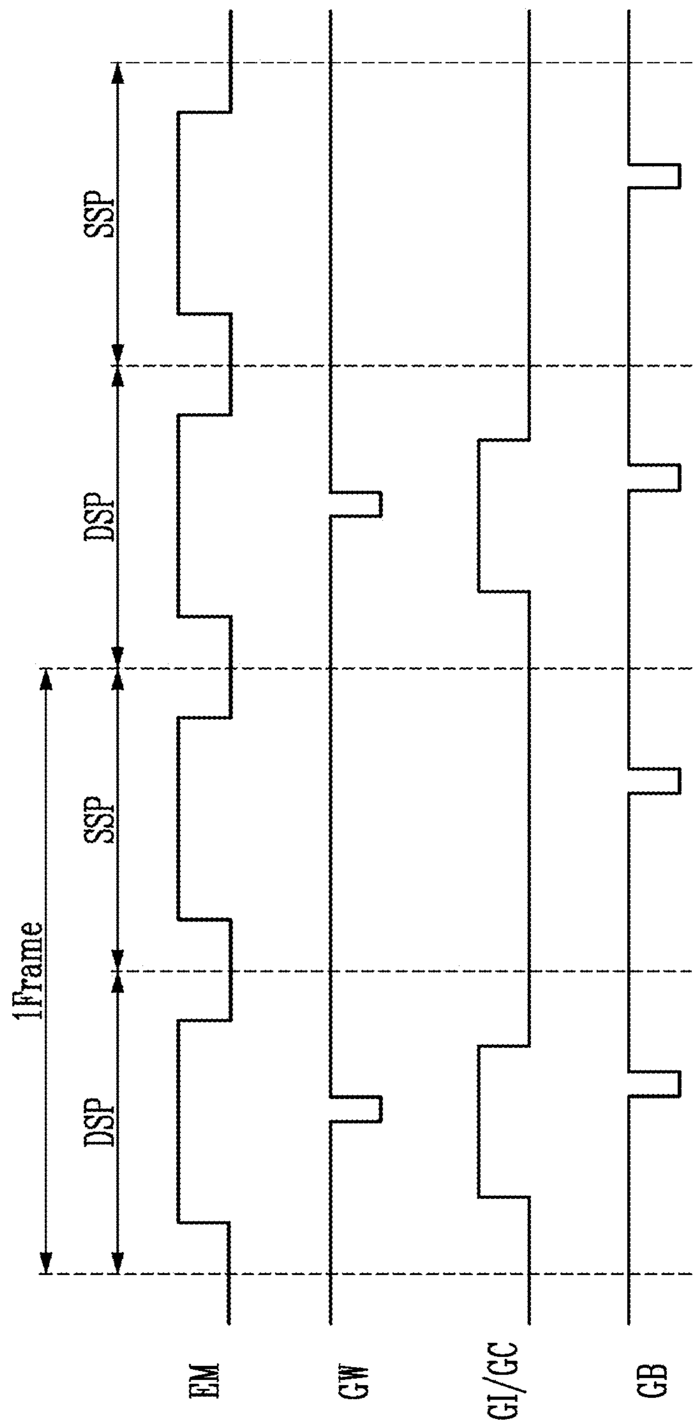


FIG. 8

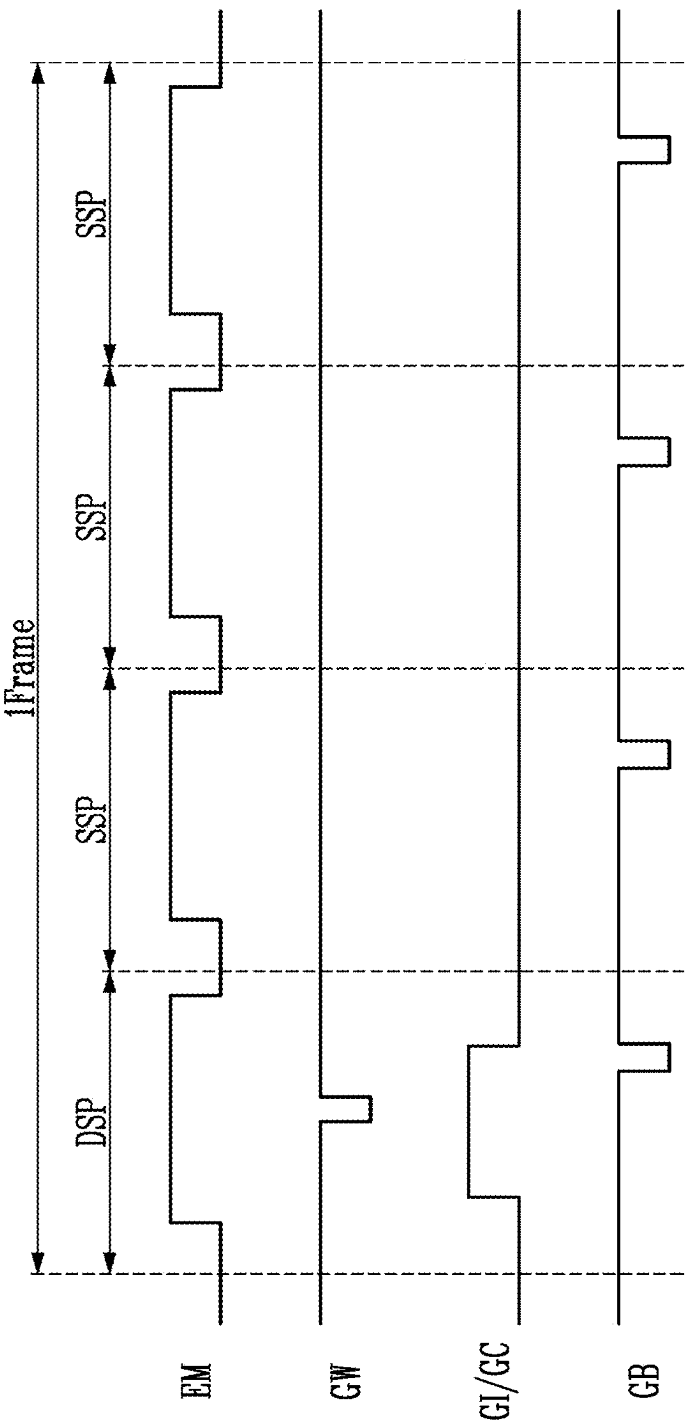


FIG. 9

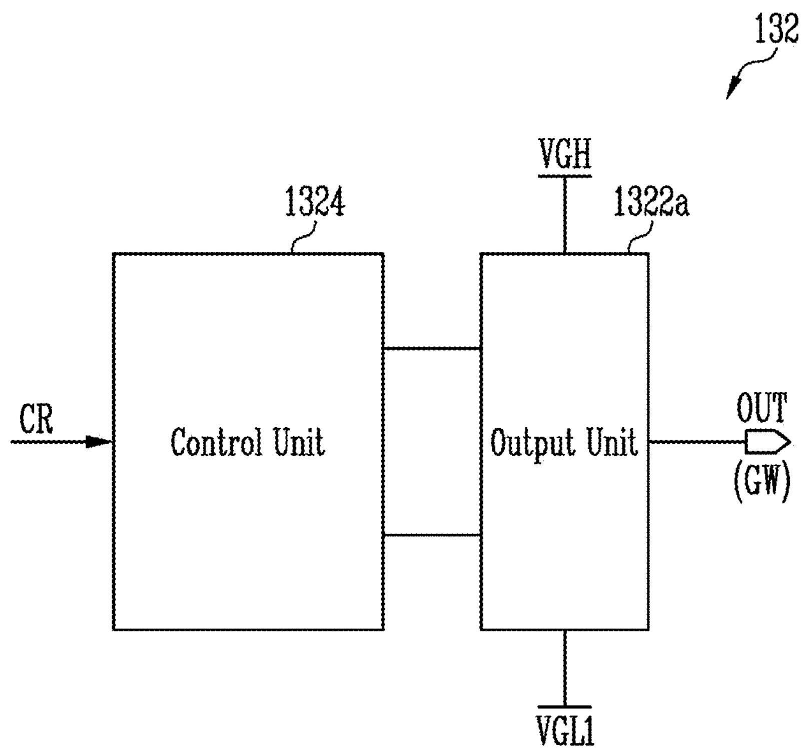


FIG. 10

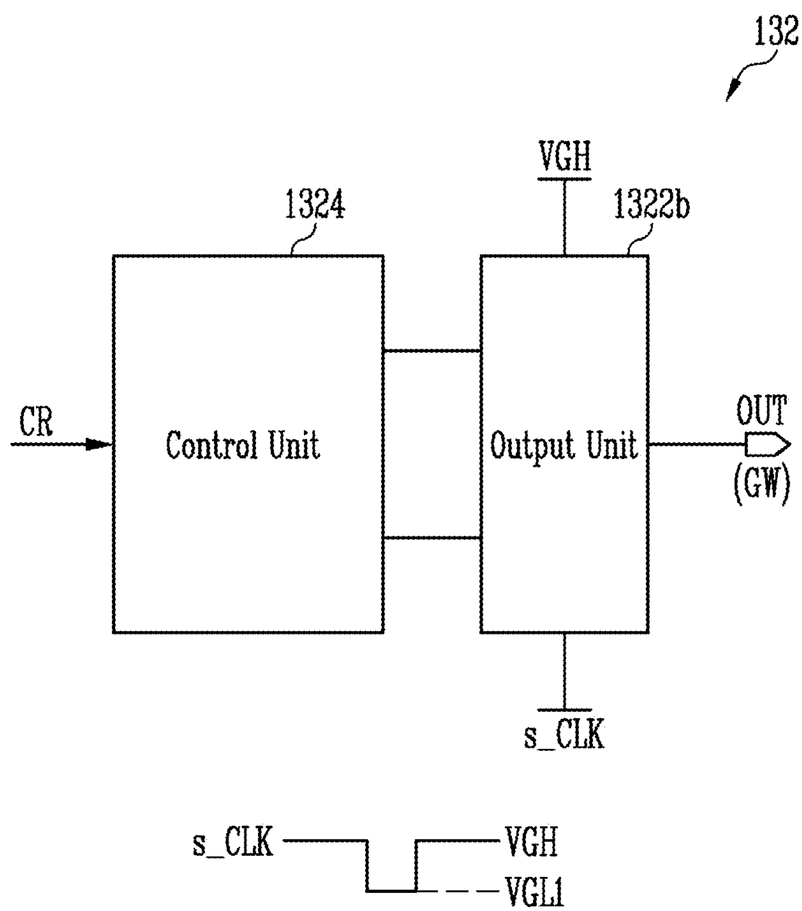
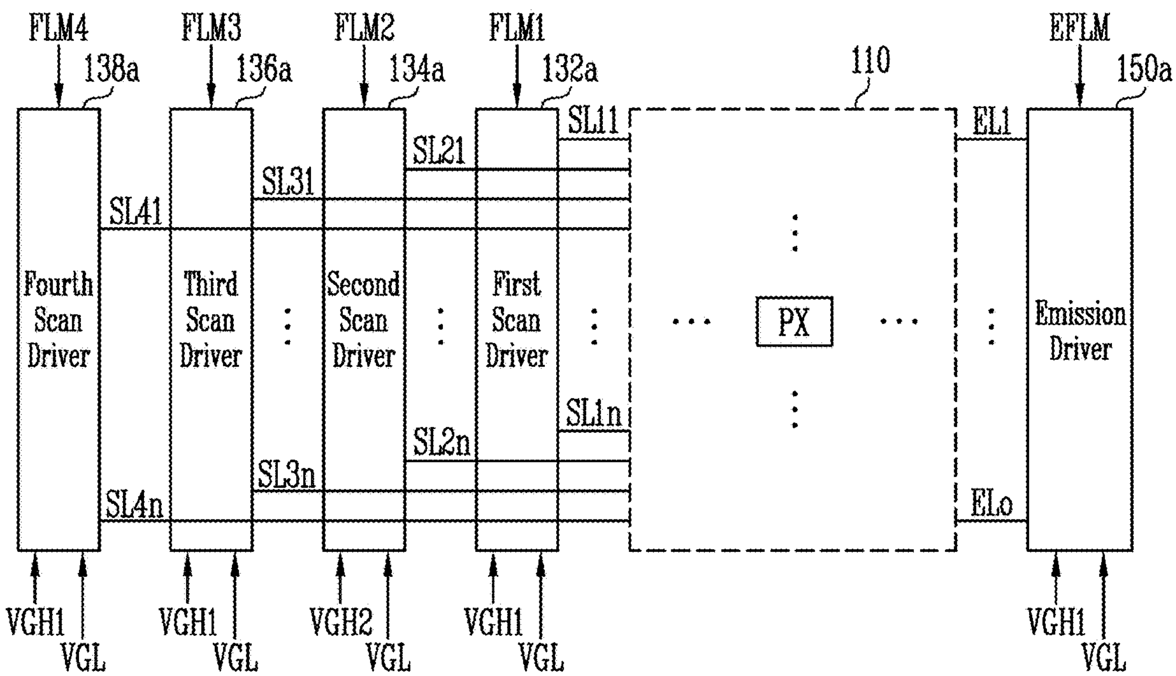

132
VGH
1324
1322b
CR
Control Unit
Output Unit
OUT
(GW)
s_CLK
s_CLK
VGH
VGL1

FIG. 11



130a: 132a, 134a, 136a, 138a

FIG. 12

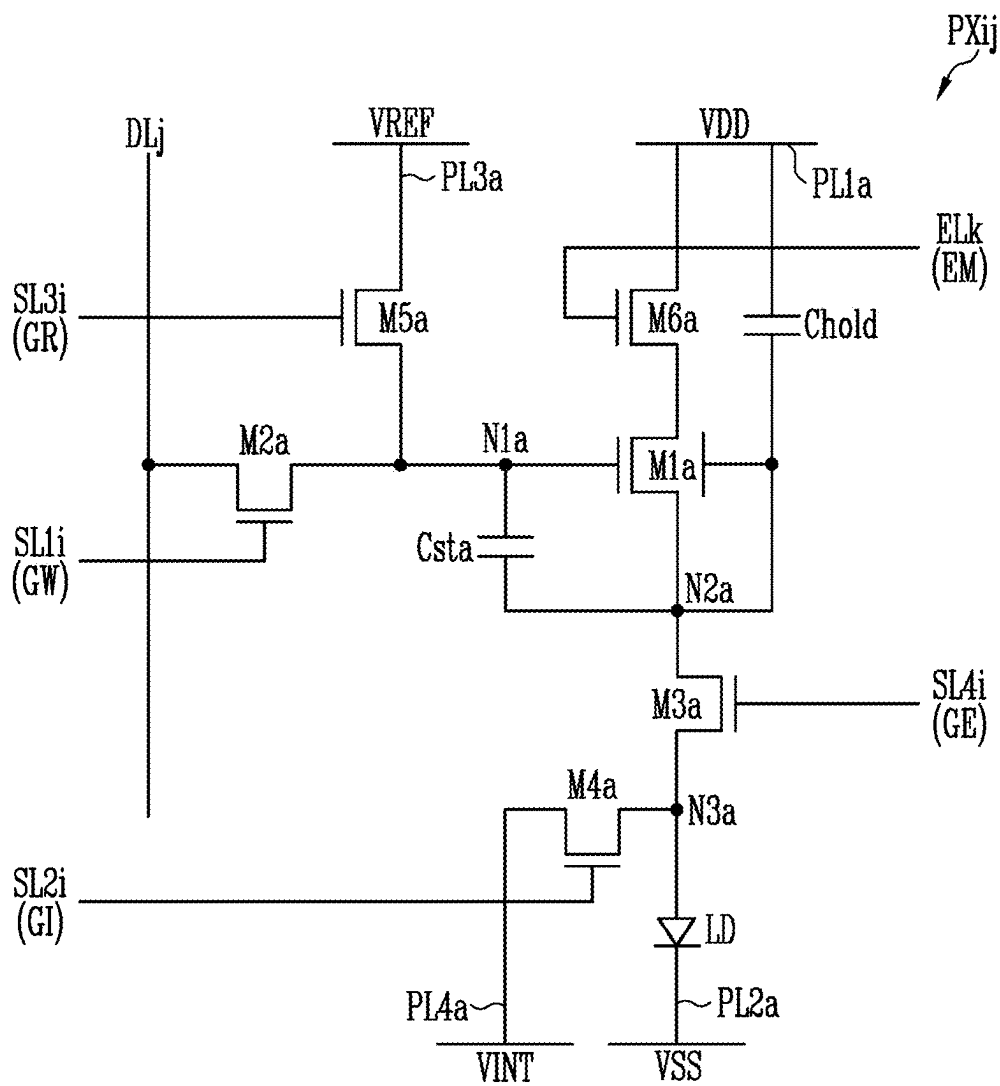

PXij
DLj
VREF
PL3a
VDD
PL1a
ELk
(EM)
SL3i
(GR)
M5a
M6a
Chold
M2a
N1a
M1a
SL1i
(GW)
Csta
N2a
M3a
SL4i
(GE)
M4a
N3a
SL2i
(GI)
LD
PL4a
PL2a
VINT
VSS

FIG. 13

<DSP>

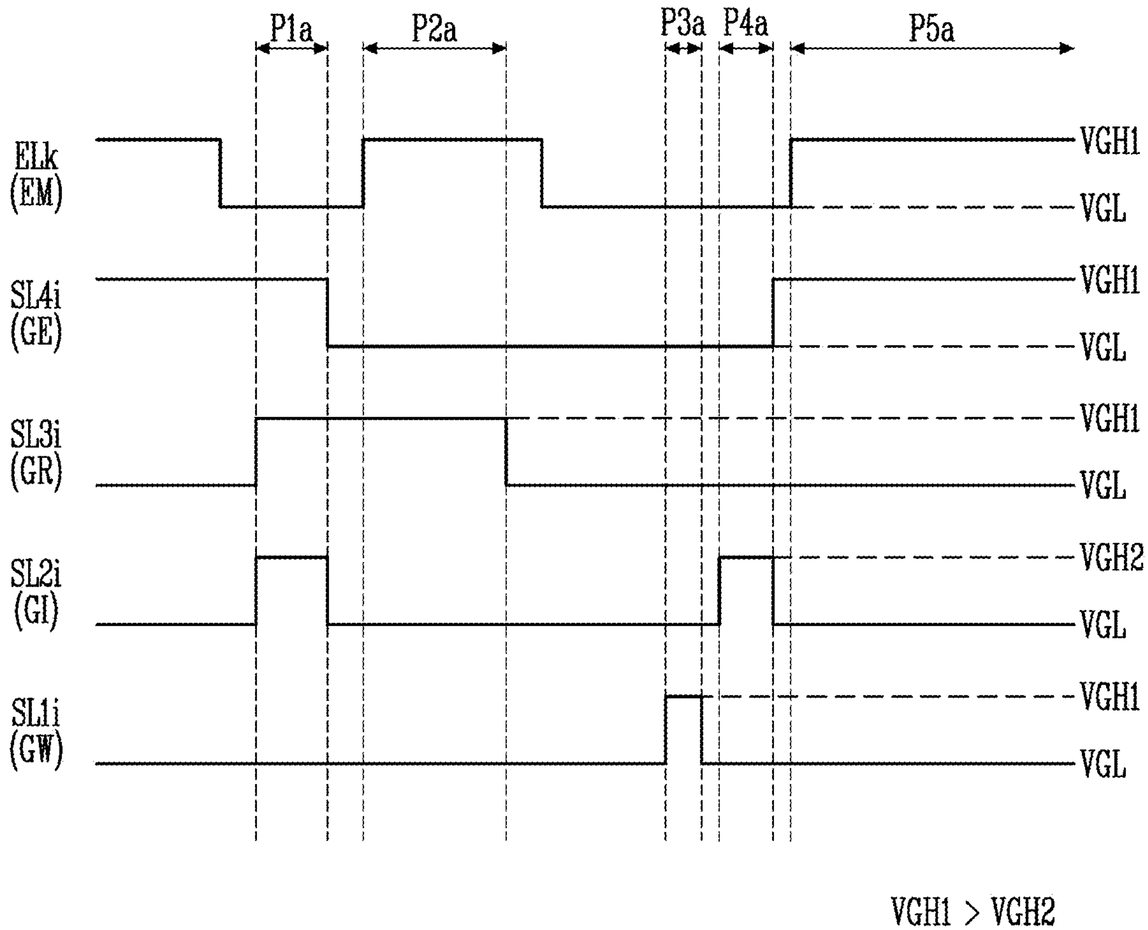

P1a
P2a
P3a
P4a
P5a
ELk
(EM)
SL4i
(GE)
SL3i
(GR)
SL2i
(GI)
SL1i
(GW)
VGH1
VGL
VGH1
VGL
VGH1
VGL
VGH2
VGL
VGH1
VGL
VGH1 > VGH2

FIG. 14

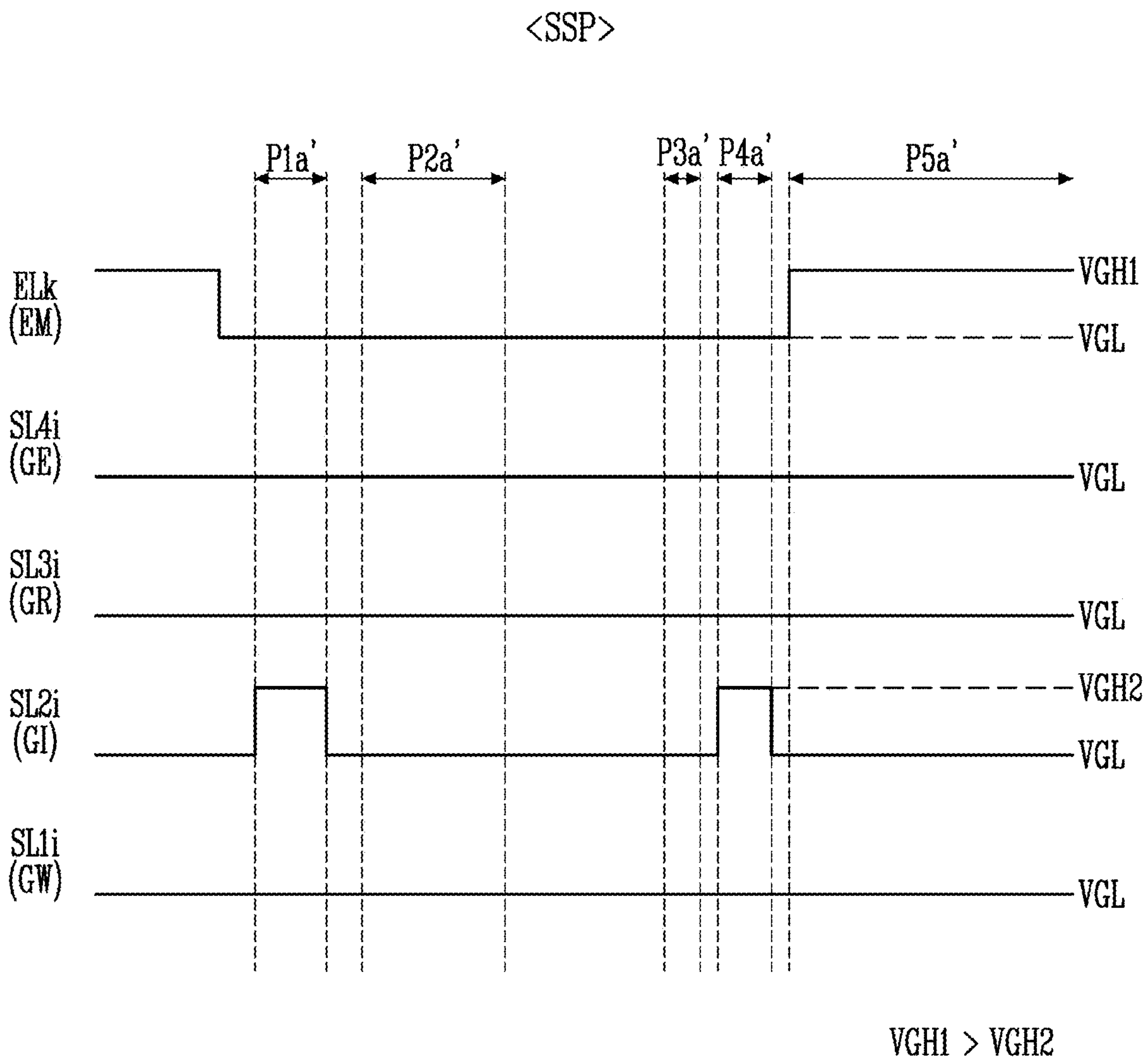
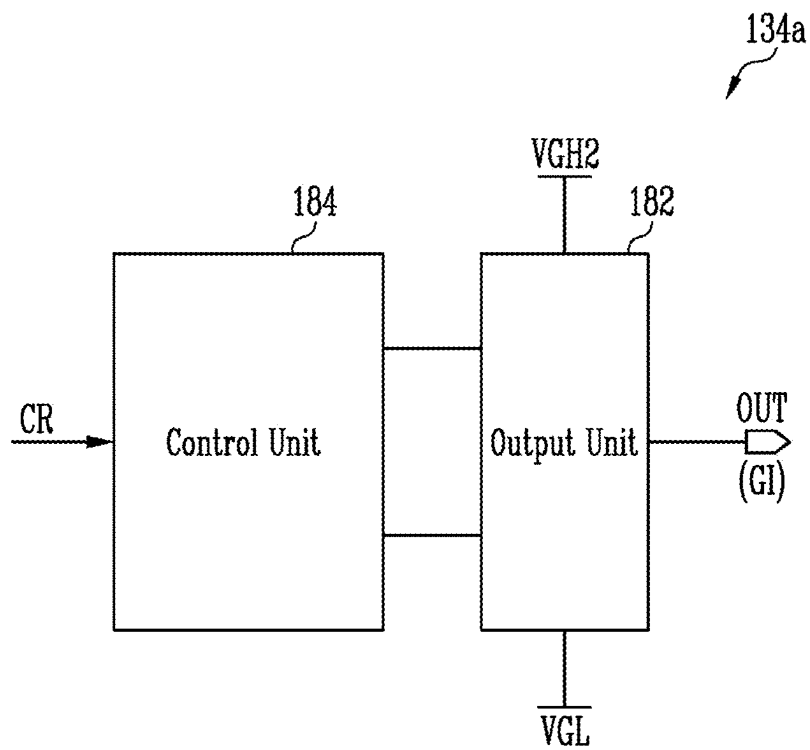

<SSP>
P1a'
P2a'
P3a'
P4a'
P5a'
ELk (EM)
SL4i (GE)
SL3i (GR)
SL2i (GI)
SL1i (GW)
VGH1
VGL
VGL
VGL
VGH2
VGL
VGL
VGH1 > VGH2

FIG. 15



DISPLAY DEVICE

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0030871, filed on Mar. 4, 2024, the disclosure of which is incorporated by reference herein in its entirety.

TECHNICAL FIELD

[0002] Embodiments of the present inventive concept relate to a display device.

DISCUSSION OF RELATED ART

[0003] As advancements are made in information technology, the utilization of a display device, such as a liquid crystal display device and an organic light emitting display device, which provides a user with information, is increasing. Such display devices may be used in a variety of devices, including a portable device. When used in a portable device, a display device having reduced power consumption is desirable.

SUMMARY

[0004] Embodiments of the present inventive concept provide a display device that can reduce power consumption by increasing a gate low voltage of a scan signal or reducing a gate high voltage of the scan signal.

[0005] A display device according to embodiments of the present inventive concept includes a pixel unit including a plurality of pixels. Each of the pixels includes a light emitting element, a driving transistor configured to control an amount of current flowing from a first power source line to a second power source line via the light emitting element in response to a voltage of a first node, a write transistor connected between a first electrode of the driving transistor and a data line and configured to be turned on or off in response to a write scan signal, and an initialization transistor connected between an anode electrode of the light emitting element and an initialization power source line and configured to be turned on or off in response to an initialization scan signal. A low voltage of the write scan signal is set to a first gate low voltage, and a low voltage of the initialization scan signal is set to a second gate low voltage that is different from the first gate low voltage.

[0006] According to an embodiment, the write transistor and the initialization transistor are P-type transistors.

[0007] According to an embodiment, a high voltage of the write scan signal and a high voltage of the initialization scan signal are set to a gate high voltage.

[0008] According to an embodiment, the first gate low voltage is a voltage higher than the second gate low voltage.

[0009] According to an embodiment, the display device further includes a data driver configured to supply a data signal to the data line, a first scan driver configured to supply the write scan signal to a write scan line, and a second scan driver configured to supply the initialization scan signal to an initialization scan line.

[0010] According to an embodiment, the first scan driver includes a plurality of stage circuits. Each of the stage circuits includes an output unit configured to generate the write scan signal using a gate high voltage and the first gate low voltage, and a control unit configured to control the output unit.

[0011] According to an embodiment, the first scan driver includes a plurality of stage circuits. Each of the stage circuits includes an output unit configured to generate the write scan signal using a gate high voltage and a clock signal, and a control unit configured to control the output unit.

[0012] According to an embodiment, a high voltage of the clock signal is set to the gate high voltage, and a low voltage of the clock signal is set to the first gate low voltage.

[0013] A display device according to embodiments of the present inventive concept includes a pixel unit including a plurality of pixels. Each of the pixels includes a light emitting element, a driving transistor configured to control an amount of current flowing from a first power source line to a second power source line via the light emitting element in response to a voltage of a first node, a write transistor connected between the first node and a data line and configured to be turned on or off in response to a write scan signal, and an initialization transistor connected between an anode electrode of the light emitting element and an initialization power source line and configured to be turned on or off in response to an initialization scan signal. A high voltage of the write scan signal is set to a first gate high voltage, and a high voltage of the initialization scan signal is set to a second gate high voltage.

[0014] According to an embodiment, the write transistor and the initialization transistor are N-type transistors.

[0015] According to an embodiment, a low voltage of the write scan signal and a low voltage of the initialization scan signal are set to a gate low voltage.

[0016] According to an embodiment, the first gate high voltage is a voltage higher than the second gate high voltage.

[0017] According to an embodiment, each of the pixels further includes a first emission transistor connected between a second electrode of the driving transistor and the anode electrode of the light emitting element and configured to be turned on or off by a first control scan signal, a second emission transistor connected between the first power source line and a first electrode of the driving transistor and configured to be turned on or off by an emission control signal, a control transistor connected between a reference power source line and the first node and configured to be turned on or off by a second control scan signal, a first capacitor connected between the first node and the second electrode of the driving transistor, and a second capacitor connected between the first power source line and the second electrode of the driving transistor. The first emission transistor, the second emission transistor, and the control transistor are N-type transistors.

[0018] According to an embodiment, a high voltage of each of the first control scan signal, the second control scan signal, and the emission control signal is set to the first gate high voltage, and a low voltage of each of the first control scan signal, the second control scan signal, and the emission control signal is set to a gate low voltage.

[0019] According to an embodiment, a high voltage of each of the first control scan signal and the emission control signal is set to the first gate high voltage, and a low voltage of each of the first control scan signal and the emission control signal is set to a gate low voltage. Further, a high voltage of the second control scan signal is set to the second gate high voltage, and a low voltage of the second control scan signal is set to the gate low voltage.

[0020] According to an embodiment, the display device further includes a data driver configured to supply a data signal to the data line, a first scan driver configured to supply the write scan signal to a write scan line, and a second scan driver configured to supply the initialization scan signal to an initialization scan line.

[0021] According to an embodiment, the second scan driver includes a plurality of stage circuits. Each of the stage circuits includes an output unit configured to generate the initialization scan signal using the second gate high voltage and a gate low voltage, and a control unit configured to control the output unit.

[0022] A display device according to embodiments of the present inventive concept includes a pixel unit including a plurality of pixels connected to a plurality of first scan lines, a plurality of second scan lines, and a plurality of data lines, a power source generator configured to generate a gate high voltage, a first gate low voltage, and a second gate low voltage using a first power source input from outside of the power source generator and a second power source having a voltage lower than the first power source, a scan driver configured to generate a first scan signal to be supplied to the first scan lines using the gate high voltage and the first gate low voltage, and configured to generate a second scan signal to be supplied to the second scan lines using the gate high voltage and the second gate low voltage, and a data driver configured to supply a data signal to the data lines.

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The above and other features of the present inventive concept will become more apparent by describing in detail embodiments thereof with reference to the accompanying drawings.

[0024] FIG. **1** is a diagram illustrating a display device according to an embodiment of the present inventive concept.

[0025] FIG. **2** is a diagram illustrating a scan driver and an emission driver according to an embodiment of the present inventive concept.

[0026] FIG. **3** is a diagram illustrating a scan driver and an emission driver according to an embodiment of the present inventive concept.

[0027] FIG. **4** is a diagram illustrating a pixel according to an embodiment of the present inventive concept.

[0028] FIG. **5** is a waveform diagram illustrating an embodiment of a method of driving the pixel of FIG. **4** during a display scan period.

[0029] FIG. **6** is a waveform diagram illustrating an embodiment of a method of driving the pixel of FIG. **4** during a self-scan period.

[0030] FIGS. **7** and **8** are diagrams illustrating an embodiment of signals supplied during an active period and a blank period.

[0031] FIG. **9** is a diagram illustrating an embodiment of a stage circuit of a first scan driver.

[0032] FIG. **10** is a diagram illustrating an embodiment of a stage circuit of the first scan driver.

[0033] FIG. **11** is a diagram illustrating a scan driver and an emission driver according to an embodiment of the present inventive concept.

[0034] FIG. **12** is a diagram illustrating a pixel according to an embodiment of the present inventive concept.

[0035] FIG. **13** is a diagram illustrating an embodiment of a method of driving the pixel of FIG. **12** during a display scan period.

[0036] FIG. **14** is a diagram illustrating an embodiment of a method of driving the pixel of FIG. **12** during a self-scan period.

[0037] FIG. **15** is a diagram illustrating an embodiment of a stage circuit of a second scan driver shown in FIG. **11**.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0038] Embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings. Like reference numerals may refer to like elements throughout the accompanying drawings.

[0039] Some embodiments are described with reference to the accompanying drawings in relation to a functional block, unit, and/or module. Those skilled in the art will understand that such a block, unit, and/or module may be physically implemented by, for example, a logic circuit, an individual component, a microprocessor, a hardwired circuit, a memory element, a line connection, and other electronic circuits, and may be formed using a semiconductor-based manufacturing technique or other manufacturing techniques. The block, unit, and/or module implemented by a microprocessor or other similar hardware may be programmed and controlled using software to perform various functions discussed herein, and may optionally be driven by firmware and/or software. In addition, each block, unit, and/or module may be implemented by dedicated hardware, or a combination of dedicated hardware that performs some functions and a processor (for example, one or more programmed microprocessors and related circuits) that performs a function different from those of the dedicated hardware. In addition, in some embodiments, the block, unit, and/or module may be physically separated into two or more individual blocks, units, and/or modules without departing from the scope of the inventive concept. In addition, in some embodiments, the block, unit and/or module may be physically combined into more complex blocks, units, and/or modules without departing from the scope of the inventive concept.

[0040] The term "connection" between two components may mean that both of an electrical connection and a physical connection are used inclusively, but the present inventive concept is not limited thereto. For example, a "connection" with reference to a circuit diagram may mean an electrical connection, and a "connection" with reference to a cross-sectional view and a plan view may mean a physical connection.

[0041] It will be understood that the terms "first," "second," "third," etc. are used herein to distinguish one element from another, and the elements are not limited by these terms. Thus, a "first" element in an embodiment may be described as a "second" element in another embodiment.

[0042] It should be understood that descriptions of features or aspects within each embodiment should typically be considered as available for other similar features or aspects in other embodiments, unless the context clearly indicates otherwise.

[0043] As used herein, the singular forms "a", "an" and "the" are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0044] Herein, when two or more elements or values are described as being substantially the same as or about equal to each other, it is to be understood that the elements or values are identical to each other, the elements or values are equal to each other within a measurement error, or if measurably unequal, are close enough in value to be functionally equal to each other as would be understood by a person having ordinary skill in the art. For example, the term "about" as used herein is inclusive of the stated value and means within an acceptable range of deviation for the particular value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the particular quantity (e.g., the limitations of the measurement system). For example, "about" may mean within one or more standard deviations as understood by one of the ordinary skill in the art. Further, it is to be understood that while parameters may be described herein as having "about" a certain value, according to embodiments, the parameter may be exactly the certain value or approximately the certain value within a measurement error as would be understood by a person having ordinary skill in the art. Other uses of these terms and similar terms to describe the relationships between components should be interpreted in a like fashion.

[0045] FIG. **1** is a diagram illustrating a display device according to an embodiment of the present inventive concept.

[0046] Referring to FIG. **1**, a display device **100** according to an embodiment of the present inventive concept may include a pixel unit **110** (or display panel), a timing controller **120** (also referred to as a timing controller circuit), a scan driver **130** (also referred to as a scan driver circuit), a data driver **140** (also referred to as a data driver circuit), an emission driver **150** (also referred to as an emission driver circuit), a power supply unit **160** (also referred to as a power supply circuit), and a power source generator **170** (also referred to as a power source generator circuit).

[0047] The display device **100** may display images at various image refresh rates (driving frequencies or screen refresh rates) depending on driving conditions. The image refresh rate may refer to the frequency at which data signals are written to a driving transistor of a pixel PX. For example, the image refresh rate may be referred to as the screen refresh rate, and may indicate the frequency at which the display screen is refreshed for 1 second.

[0048] In an embodiment, in correspondence with the image refresh rate, an output frequency of the data driver **140** for one horizontal line (for example, pixels PX connected to the same scan line may be classified into one horizontal line (or pixel row)) and/or an output frequency of a first scan driver **132** outputting a first scan signal (or write scan signal) can be determined. For example, the image refresh rate for driving a moving image may be a frequency of about 60 Hz or higher (for example, about 120 Hz, about 240 Hz, etc.).

[0049] For example, the display device **100** may display images in response to various image refresh rates from about 1 Hz to about 240 Hz. However, this is only an example, and the display device **100** may display images at an image refresh rate of about 240 Hz or higher (for example, about 480 Hz) according to embodiments.

[0050] The pixel unit **110** may include pixels PX connected to first scan lines SL**11**, SL**12**, . . . , and SL**1***n*, second scan lines SL**21**, SL**22**, . . . , and SL**2***n*, third scan lines SL**31**, SL**32**, . . . , and SL**3***n*, fourth scan lines SL**41**, SL**42**, . . . , and SL**4***n*, data lines DL**1**, DL**2**, . . . , and DLm, emission control lines EL**1**, EL**2**, . . . , and ELo, and power source lines PL**1**, PL**2**, PL**3**, PL**4**, and PL**5**, where n, m, and o are positive integers greater than or equal to 2.

[0051] As an example, a pixel PXij (see FIG. **4**) located on an i-th horizontal line (or pixel row) and a j-th vertical line (or pixel column) may be connected to an i-th first scan line SL**1***i*, an i-th second scan line SL**2***i*, an i-th third scan line SL**3***i*, an i-th fourth scan line SL**4***i*, a k-th emission control line ELk, and a j-th data line DLj, where i is a positive integer less than or equal to n, j is a positive integer less than or equal to m, and k is a positive integer less than or equal to o. Here, k is a number equal to or smaller than i. As an example, when each of the emission control lines EL**1** to ELo is connected to pixels PX located on one horizontal line, k may be the same number as i. As an example, when each of the emission control lines EL**1** to ELo is connected to pixels PX located on two or more horizontal lines, k may be a number smaller than i.

[0052] The pixels PX may be selected in units of horizontal lines when an enable first scan signal is supplied to the first scan lines SL**11** to SL**1***n*, and each of the pixels PX selected by the enable first scan signal may receive a data signal from a data line (any one of DL**1** to DLm) connected thereto. The pixels PX that receive the data signal may generate light with a predetermined luminance in response to a voltage of the data signal.

[0053] The scan driver **130** may receive a scan driving signal SCS from the timing controller **120**. The scan driving signal SCS may include at least one scan start signal and clock signals utilized to drive the scan driver **130**. The scan driver **130** may generate an enable first scan signal, an enable second scan signal, an enable third scan signal, and an enable fourth scan signal while shifting the scan start signal in response to a clock signal.

[0054] The data driver **140** may receive output data Dout and a data driving signal DCS from the timing controller **120**. The data driving signal DCS may include a sampling signal and/or timing signals utilized to drive the data driver **140**. The data driver **140** may generate the data signal based on the data driving signal DCS and the output data Dout. As an example, the data driver **140** may generate an analog data signal based on a grayscale of the output data Dout. The data driver **140** may supply the data signal in units of one horizontal period.

[0055] The emission driver **150** may receive an emission driving signal ECS from the timing controller **120**. The emission driving signal ECS may include an emission start signal and clock signals utilized to drive the emission driver **150**. The emission driver **150** may generate a disable emission control signal EM while shifting the emission start signal in response to a clock signal.

[0056] The timing controller **120** may receive input data Din and a timing control signal TCS from a host system through an interface. As an example, the timing controller **120** may receive the input data Din and the timing control signal TCS from at least one of a graphics processing unit (GPU), a central processing unit (CPU), and an application processor (AP) included in the host system. The timing control signal TCS may include various signals including a clock signal.

[0057] The timing controller **120** may generate the scan driving signal SCS, the data driving signal DCS, and the emission driving signal ECS based on the timing control signal TCS. The scan driving signal SCS, the data driving signal DCS, and the emission driving signal ECS may be supplied to the scan driver **130**, the data driver **140**, and the emission driver **150**, respectively.

[0058] The timing controller **120** may rearrange the input data Din to match the specifications of the display device **100**. In addition, the timing controller **120** may correct the input data Din to generate the output data Dout and supply the output data Dout to the data driver **140**. In an embodiment, the timing controller **120** may correct the input data Din in response to optical measurement results measured during a process.

[0059] The power supply unit **160** may generate various power sources utilized to drive the display device **100**. As an example, the power supply unit **160** may generate a first driving power source VDD, a second driving power source VSS, a first initialization power source Vint**1**, a second initialization power source Vint**2**, and a bias power source Vbias.

[0060] The first driving power source VDD may be a power source that supplies driving current to the pixels PX. The second driving power source VSS may be a power source that receives driving current from the pixels PX. During a period in which the pixels PX are set to emit light, the first driving power source VDD may be set to a voltage higher than the second driving power source VSS.

[0061] The first initialization power source Vint**1** may be a power source that initializes a gate electrode of a driving transistor included in each of the pixels PX. The first initialization power source Vint**1** may be set to a voltage lower than the data signal. The second initialization power source Vint**2** may be a power source that initializes a first electrode (or anode electrode) of a light emitting element LD included in each of the pixels PX. The second initialization power source Vint**2** may be set to a voltage at which the light emitting element LD turns off. The bias power source Vbias may be a power source for applying an on bias voltage to the driving transistor included in each of the pixels PX.

[0062] The first driving power source VDD generated by the power supply unit **160** may be supplied to a first power source line PL**1**, the second driving power source VSS may be supplied to a second power source line PL**2**, the first initialization power source Vint**1** may be supplied to a third power source line PL**3**, the second initialization power source Vint**2** may be supplied to a fourth power source line PL**4**, and the bias power source Vbias may be supplied to a fifth power source line PL**5**. The first power source line PL**1**, the second power source line PL**2**, the third power source line PL**3**, the fourth power source line PL**4**, and the fifth power source line PL**5** may be commonly connected to the pixels PX, but embodiments of the present inventive concept are not limited thereto.

[0063] In an embodiment, the first power source line PL**1** may be composed of a plurality of power source lines, and the plurality of power source lines may be connected to different pixels PX. In an embodiment, the second power source line PL**2** may be composed of a plurality of power source lines, and the plurality of power source lines may be connected to different pixels PX. In an embodiment, the third power source line PL**3** may be composed of a plurality of power source lines, and the plurality of power source lines may be connected to different pixels PX. In an embodiment, the fourth power source line PL**4** may be composed of a plurality of power source lines, and the plurality of power source lines may be connected to different pixels PX. In an embodiment, the fifth power source line PL**5** may be composed of a plurality of power source lines, and the plurality of power source lines may be connected to different pixels PX. That is, in an embodiment of the present inventive concept, the pixels PX may be connected to one of first power source lines PL**1**, one of second power source lines PL**2**, one of third power source lines PL**3**, one of fourth power source lines PL**4**, and one of fifth power line PL**5**.

[0064] The power source generator **170** may generate a gate high voltage VGH, a first gate low voltage VGL**1**, and a second gate low voltage VGL**2** using a first power source input from outside of the power source generator **170**. At least two voltages among the gate high voltage VGH, the first gate low voltage VGL**1**, and the second gate low voltage VGL**2** may be supplied to the scan driver **130** and/or the emission driver **150**. The scan driver **130** and/or the emission driver **150** may generate a scan signal and an emission control signal using voltages supplied thereto (for example, at least two voltages among the gate high voltage VGH, the first gate low voltage VGL**1**, and the second gate low voltage VGL**2**).

[0065] The gate high voltage VGH, the first gate low voltage VGL**1**, and the second gate low voltage VGL**2** may be supplied to the timing controller **120**. The timing controller **120** may generate a clock signal to be supplied to the scan driver **130** and/or the emission driver **150** using the gate high voltage VGH, the first gate low voltage VGL**1**, and the second gate low voltage VGL**2**. As an example, the timing controller **120** may generate a clock signal s_CLK (see FIG. **10**) using the gate high voltage VGH and the first gate low voltage VGL**1**.

[0066] In addition, the timing controller **120** may generate a second clock signal using the gate high voltage VGH and the second gate low voltage VGL**2**. Here, the first gate low voltage VGL**1** may be set to a voltage higher than the second gate low voltage VGL**2**. In addition, the gate high voltage VGH may be set to a voltage higher than the first gate low voltage VGL**1** and the second gate low voltage VGL**2**.

[0067] In an embodiment of the present inventive concept, the display device **100** may include, for example, a flat panel display device, a curved display device in which a portion of the pixel unit **110** is curved, a flexible display device that can be partially folded or bent, or a stretchable display device that can be partially stretched.

[0068] In an embodiment of the present inventive concept, the display device **100** may be a device that displays a moving image or a still image and may include portable electronic devices such as, for example, a mobile phone, a smartphone, a tablet personal computer, a smartwatch, a watch phone, a mobile communication terminal, an electronic notebook, an e-book, a portable multimedia player (PMP), a navigation system, and an ultra-mobile PC (UMPC). In an embodiment of the present inventive concept, the display device **100** may include electronic devices such as, for example, a television, a laptop, a monitor, a billboard, and an Internet of Things (IoT) device.

[0069] FIG. **2** is a diagram illustrating a scan driver and an emission driver according to an embodiment of the present inventive concept.

[0070] Referring to FIG. **2**, the scan driver **130** according to an embodiment of the present inventive concept may include a first scan driver **132**, a second scan driver **134**, a third scan driver **136**, and a fourth scan driver **138**. Depending on the design, at least some of the scan drivers **132**, **134**, **136**, and **138** may be integrated into one driving circuit, module, etc.

[0071] The first scan driver **132** may receive a first scan start signal FLM**1** and generate an enable first scan signal while shifting the first scan start signal FLM**1** in response to a clock signal. The first scan driver **132** may sequentially supply the enable first scan signal to the first scan lines SL**11** to SL**1**n. In an embodiment, the first scan driver **132** may supply the enable first scan signal during a display scan period of one frame.

[0072] The first scan driver **132** may generate a first scan signal using the gate high voltage VGH and the first gate low voltage VGL**1**. As an example, a gate-off voltage of the first scan signal supplied to a P-type transistor may be set to the gate high voltage VGH, and a gate-on voltage may be set to the first gate low voltage VGL**1**. The enable first scan signal may be set to the first gate low voltage VGL**1** so that the P-type transistor can be turned on, and a disable first scan signal may be set to the gate high voltage VGH so that the P-type transistor can be turned off. Additionally, the first gate low voltage VGL**1** may be supplied to the first scan driver **132** in the form of a pulse.

[0073] The second scan driver **134** may receive a second scan start signal FLM**2** and generate an enable second scan signal while shifting the second scan start signal FLM**2** in response to a clock signal. The second scan driver **134** may sequentially supply the enable second scan signal to the second scan lines SL**21** to SL**2**n. In an embodiment, the second scan driver **134** may supply the enable second scan signal during the display scan period of one frame.

[0074] The second scan driver **134** may generate a second scan signal using the gate high voltage VGH and the second gate low voltage VGL**2**. As an example, a gate-off voltage of the second scan signal supplied to an N-type transistor may be set to the second gate low voltage VGL**2**, and a gate-on voltage may be set to the gate high voltage VGH. The enable second scan signal may be set to the gate high voltage VGH so that the N-type transistor can be turned on, and a disable second scan signal may be set to the second gate low voltage VGL**2** so that the N-type transistor can be turned off.

[0075] The third scan driver **136** may receive a third scan start signal FLM**3** and generate an enable third scan signal while shifting the third scan start signal FLM**3** in response to a clock signal. The third scan driver **136** may sequentially supply the enable third scan signal to the third scan lines SL**31** to SL**3**n. In an embodiment, the third scan driver **136** may supply the enable third scan signal during the display scan period of one frame.

[0076] The third scan driver **136** may generate a third scan signal using the gate high voltage VGH and the second gate low voltage VGL**2**. As an example, a gate-off voltage of the third scan signal supplied to an N-type transistor may be set to the second gate low voltage VGL**2**, and a gate-on voltage may be set to the gate high voltage VGH. The enable third scan signal may be set to the gate high voltage VGH so that the N-type transistor can be turned on, and a disable third scan signal may be set to the second gate low voltage VGL**2** so that the N-type transistor can be turned off.

[0077] The fourth scan driver **138** may receive a fourth scan start signal FLM**4** and generate an enable fourth scan signal while shifting the fourth scan start signal FLM**4** in response to a clock signal. The fourth scan driver **138** may sequentially supply the enable fourth scan signal to the fourth scan lines SL**41** to SL**4**n.

[0078] In an embodiment, the fourth scan driver **138** may supply the enable fourth scan signal during a display scan period and self-scan period of one frame. For example, the fourth scan driver **138** may perform scanning once during the display scan period (that is, supply at least one enable fourth scan signal), and perform scanning at least once according to the image refresh rate during the self-scan period. When the image refresh rate is reduced (that is, the frame length becomes longer), the number of repetitions of the operation in which the fourth scan driver **138** supplies the enable fourth scan signal to each of the fourth scan lines SL**41** to SL**4**n within one frame period may be increased.

[0079] The fourth scan driver **138** may generate a fourth scan signal using the gate high voltage VGH and the second gate low voltage VGL**2**. As an example, a gate-off voltage of the fourth scan signal supplied to a P-type transistor may be set to the gate high voltage VGH, and a gate-on voltage may be set to the second gate low voltage VGL**2**. The enable fourth scan signal may be set to the second gate low voltage VGL**2** so that the P-type transistor can be turned on, and a disable fourth scan signal may be set to the gate high voltage VGH so that the P-type transistor can be turned off.

[0080] FIG. **2** shows the first scan driver **132**, the second scan driver **134**, the third scan driver **136**, and the fourth scan driver **138** connected to the first scan line SL**1**, the second scan line SL**2**, the third scan line SL**3**, and the fourth scan line SL**4**, respectively, but embodiments of the present inventive concept are not limited thereto. For example, according to embodiments, at least two scan lines (at least two of SL**1**, SL**2**, SL**3**, and SL**4**) among the first scan line SL**1**, the second scan line SL**2**, the third scan line SL**3**, and the fourth scan line SL**4** may be driven by one scan driver.

[0081] As an example, the second scan line SL**2** and the third scan line SL**3** may be driven by one scan driver. In this case, one of the second scan driver **134** and the third scan driver **136** shown in FIG. **2** may be omitted.

[0082] The emission driver **150** may receive an emission start signal EFLM and generate a disable emission control signal EM while shifting the emission start signal EFLM in response to a clock signal. The emission driver **150** may sequentially supply the disable emission control signal EM to the emission control lines EL**1** to ELo.

[0083] The emission driver **150** may generate an emission control signal using the gate high voltage VGH and the second gate low voltage VGL**2**. As an example, a gate-off voltage of the emission control signal supplied to a P-type transistor may be set to the gate high voltage VGH, and a gate-on voltage may be set to the second gate low voltage VGL**2**. The disable emission control signal may be set to the gate high voltage VGH so that the P-type transistor can be turned off, and an enable emission control signal may be set to the second gate low voltage VGL**2** so that the P-type transistor can be turned on.

[0084] In an embodiment, the emission driver **150** may supply the disable emission control signal during the display scan period and self-scan period of one frame. For example, the emission driver **150** may perform scanning once during the display scan period, and perform scanning at least once according to the image refresh rate during the self-scan period. When the image refresh rate is reduced (that is, the frame length becomes longer), the number of repetitions of the operation in which the emission driver **150** supplies the disable emission control signal to each of the emission control lines EL**1** to ELo within one frame period may be increased.

[0085] FIG. **2** shows an embodiment in which the first to fourth scan drivers **132** to **138** are disposed on one side of the pixel unit **110**, and the emission driver **150** is disposed on the other side of the pixel unit **110**, but embodiments of the present inventive concept are not limited thereto.

[0086] FIG. **3** is a diagram illustrating a scan driver and an emission driver according to an embodiment of the present inventive concept. In FIG. **3**, compared to FIG. **2**, the functions may be substantially the same except that the positions of the drivers are changed. Accordingly, for convenience of explanation, descriptions that overlap with those of FIG. **2** will be omitted.

[0087] Referring to FIG. **3**, the first scan driver **132** may include sub-first scan drivers **132***a* and **132***b*. The sub-first scan driver **132***a* may be disposed on one side of the pixel unit **110**, and the sub-first scan driver **132***b* may be disposed on the opposite side of the pixel unit **110**.

[0088] The third scan driver **136** and the emission driver **150** may be disposed on one side of the pixel unit **110**, and the second scan driver **134** and the fourth scan driver **138** may be disposed on the opposite side of the pixel unit **110**. In an embodiment of the present inventive concept, the first scan driver **132**, the second scan driver **134**, the third scan driver **136**, the fourth scan driver **138**, and the emission driver **150** may be arranged in various shapes.

[0089] FIG. **4** is a diagram illustrating a pixel according to an embodiment of the present inventive concept. FIG. **4** shows a pixel located on an i-th horizontal line and a j-th vertical line.

[0090] Referring to FIG. **4**, a pixel PXij according to an embodiment of the present inventive concept may be connected to corresponding signal lines SL**1***i*, SL**2***i*, SL**3***i*, SL**4***i*, ELk, and DLj. For example, the pixel PXij may be connected to an i-th first scan line SL**1***i*, an i-th second scan line SL**2***i*, an i-th third scan line SL**3***i*, an i-th fourth scan line SL**4***i*, a k-th emission control line ELk, and a j-th data line DLj. In an embodiment, the pixel PXij may be further connected to a first power source line PL**1**, a second power source line PL**2**, a third power source line PL**3**, a fourth power source line PL**4**, and a fifth power source line PL**5**.

[0091] The pixel PXij according to an embodiment of the present inventive concept may include a light emitting element LD and a pixel circuit that controls the amount of current supplied to the light emitting element LD.

[0092] The light emitting element LD may be connected between the first power source line PL**1** and the second power source line PL**2**. As an example, a first electrode (or anode electrode) of the light emitting element LD may be electrically connected to the first power source line PL**1** via a seventh transistor M**7**, a third node N**3**, a first transistor M**1**, a second node N**2**, and a sixth transistor M**6**. A second electrode (or cathode electrode) of the light emitting element LD may be electrically connected to the second power source line PL**2**. The light emitting element LD may generate light with a predetermined luminance in response to the amount of current supplied from the first power source line PL**1** to the second power source line PL**2** via the pixel circuit.

[0093] The light emitting element LD may be, for example, an organic light emitting diode or an inorganic light emitting diode, such as, for example, a micro LED (light emitting diode) or a quantum dot light emitting diode. In addition, the light emitting element LD may be an element composed of a combination of organic and inorganic materials. FIG. **4** shows the pixel PXij including a single light emitting element LD. However, embodiments of the present inventive concept are not limited thereto. For example, in an embodiment, the pixel PXij may include a plurality of light emitting elements LD, and the plurality of light emitting elements LD may be connected in series, in parallel, or in series and parallel.

[0094] The pixel circuit includes the first transistor M**1**, a second transistor M**2**, a third transistor M**3**, a fourth transistor M**4**, a fifth transistor M**5**, the sixth transistor M**6**, the seventh transistor M**7**, an eighth transistor M**8**, and a storage capacitor Cst.

[0095] A first electrode of the first transistor M**1** (or driving transistor) may be connected to the second node N**2**, and a second electrode may be connected to the third node N**3**. In addition, a gate electrode of the first transistor M**1** may be connected to a first node N**1**. The first transistor M**1** may control the amount of current supplied from a first driving power source VDD to a second driving power source VSS via the light emitting element LD in response to a voltage of the first node N**1**.

[0096] The second transistor M**2** (or write transistor) may be connected between a data line DLj and the second node N**2** (or the first electrode of the first transistor M**1**). In addition, a gate electrode of the second transistor M**2** may be electrically connected to a first scan line SL**1***i* (or write scan line). The second transistor M**2** may be turned on when an enable first scan signal GW (or write scan signal) is supplied to the first scan line SL**1***i* to electrically connect to the data line DLj and the second node N**2**.

[0097] A first electrode of the third transistor M**3** may be connected to the first node N**1**, and a second electrode may be electrically connected to the third power source line PL**3**. In addition, a gate electrode of the third transistor M**3** may be electrically connected to a third scan line SL**3***i*. The third transistor M**3** may be turned on when an enable third scan signal GI is supplied to the third scan line SL**3***i* to supply a voltage of a first initialization power source Vint**1** to the first node N**1**.

[0098] The fourth transistor M**4** may be connected between the first node N**1** and the third node N**3**. In addition, a gate electrode of the fourth transistor M**4** may be electrically connected to a second scan line SL**2***i*. The fourth transistor M**4** may be turned on when an enable second scan signal GC is supplied to the second scan line SL**2***i* to electrically connect the first node N**1** and the third node N**3**. That is, when the fourth transistor M**4** is turned on, the first transistor M**1** may be connected in the form of a diode.

[0099] A first electrode of the fifth transistor M**5** (or initialization transistor) may be connected to the first electrode of the light emitting element LD, and a second electrode may be electrically connected to the fourth power source line PL**4** (or initialization power source line). In addition, a gate electrode of the fifth transistor M**5** may be electrically connected to a fourth scan line SL**4***i* (or initialization scan line). The fifth transistor M**5** may be turned on when an enable fourth scan signal GB (or initialization scan signal) is supplied to the fourth scan line SL**4***i* to supply a voltage of a second initialization power source Vint**2** to the first electrode of the light emitting element LD.

[0100] When the voltage of the second initialization power source Vint**2** is supplied to the first electrode of the light emitting element LD, a parasitic capacitor of the light emitting element LD may be discharged. As the residual voltage charged in the parasitic capacitor of the light emitting element LD is discharged (or removed), unintentional weak light emitting can be prevented. Accordingly, the black expression ability of the pixel PXij can be improved.

[0101] A first electrode of the sixth transistor M**6** may be electrically connected to the first power source line PL**1**, and a second electrode may be connected to the second node N**2**. In addition, a gate electrode of the sixth transistor M**6** may be electrically connected to an emission control line ELk. The sixth transistor M**6** may be turned off when a disable emission control signal EM is supplied to the emission control line ELk and may be turned on when an enable emission control signal EM is supplied.

[0102] The seventh transistor M**7** may be connected between the third node N**3** and the first electrode of the light emitting element LD. In addition, a gate electrode of the seventh transistor M**7** may be electrically connected to the emission control line ELk. The seventh transistor M**7** may be turned off when the disable emission control signal EM is supplied to the emission control line ELk and may be turned on when the enable emission control signal EM is supplied.

[0103] A first electrode of the eighth transistor M**8** may be electrically connected to the fifth power source line PL**5** (or bias power source line), and a second electrode may be connected to the second node N**2**. In addition, a gate electrode of the eighth transistor M**8** may be electrically connected to the fourth scan line SL**4***i*. The eighth transistor M**8** may be turned on when the enable fourth scan signal GB is supplied to the fourth scan line SL**4***i* to electrically connect the fifth power source line PL**5** and the second node N**2**.

[0104] The storage capacitor Cst may be connected between the first power source line PL**1** and the first node N**1**. The storage capacitor Cst may store a voltage applied to the first node N**1**.

[0105] In an embodiment, the first transistor M**1**, the second transistor M**2**, the fifth transistor M**5**, the sixth transistor M**6**, the seventh transistor M**7**, and the eighth transistor M**8** may be composed of polysilicon semiconductor transistors. For example, the first transistor M**1**, the second transistor M**2**, the fifth transistor M**5**, the sixth transistor M**6**, the seventh transistor M**7**, and the eighth transistor M**8** may include a polysilicon semiconductor layer formed through a low temperature poly-silicon (LTPS) process as an active layer (channel). In addition, the first transistor M**1**, the second transistor M**2**, the fifth transistor M**5**, the sixth transistor M**6**, the seventh transistor M**7**, and the eighth transistor M**8** may be P-type transistors (for example, PMOS transistors). Accordingly, a gate-on voltage that turns on the first transistor M**1**, the second transistor M**2**, the fifth transistor M**5**, the sixth transistor M**6**, the seventh transistor M**7**, and the eighth transistor M**8** may be a logic low level. The polysilicon semiconductor transistors may provide a fast response time, which can be applied to switching elements that operate with fast switching.

[0106] In an embodiment, the third transistor M**3** and fourth transistor M**4** may be composed of oxide semiconductor transistors. For example, the third transistor M**3** and the fourth transistor M**4** may be N-type oxide semiconductor transistors (for example, NMOS transistors) and may include an oxide semiconductor layer as an active layer. Accordingly, a gate-on voltage that turns on the third transistor M**3** and the fourth transistor M**4** may be a logic high level.

[0107] The oxide semiconductor transistors may be manufactured through a low-temperature process and may have lower charge mobility than polysilicon semiconductor transistors. That is, the oxide semiconductor transistors may have excellent off-current characteristics. Therefore, when the third transistor M**3** and the fourth transistor M**4** are composed of oxide semiconductor transistors, leakage current from the first node N**1** due to low-frequency driving can be minimized, and display quality can be improved accordingly.

[0108] A gate-on voltage (or low voltage) of a fourth scan signal GB may be set to a lower voltage compared to the second initialization power source Vint**2**. As an example, when the second initialization power source Vint**2** is set to about −4.5V, the gate-on voltage may be set to about −7V or less in consideration of a threshold voltage (for example, about −2.5V) of the fifth transistor M**5**. Here, the gate-on voltage of the fourth scan signal GB may be set to the second gate low voltage VGL**2**. Accordingly, the second gate low voltage VGL**2** may be set to −about 7V or less.

[0109] When the second gate low voltage VGL**2** is set to a low voltage of the fourth scan signal GB, low voltages of the remaining transistors M**3**, M**4**, M**6**, and M**7** may also be set to the second gate low voltage VGL**2**.

[0110] In an embodiment of the present inventive concept, a low voltage of a first scan signal GW may be set to the first gate low voltage VGL**1**, which is a voltage higher than the second gate low voltage VGL**2**. A voltage of the data signal supplied to the data line DLj may be set to about 1.5V to about 7V. In this case, considering a threshold voltage (for example, about-2.5V) of the second transistor M**2**, a gate-on voltage of the second transistor M**2** may be set to about-1.0V.

[0111] That is, the data signal supplied to the data line DLj may be set to a positive voltage. Accordingly, a low voltage of the second transistor M**2** may be set to the first gate low voltage VGL**1**, which is higher than the second gate low voltage VGL**2**. In this case, the voltage swing width of the first scan signal GW may be narrowed, and thus, power consumption can be reduced.

[0112] In general, power consumption of the first scan driver **132** can be expressed as Equation 1 below.

$$P = CV^2 f \qquad \text{[Equation 1]}$$

[0113] In Equation 1, P refers to power consumption, C refers to capacitance, V refers to voltage, and f refers to frequency. In an embodiment of the present inventive concept, the low voltage of the first scan signal GW may be set to the first gate low voltage VGL**1**. In this case, a voltage (V) range can be reduced compared to a case in which the low voltage of the first scan signal GW is set to the second gate low voltage VGL**2**. Accordingly, power consumption can be reduced.

[0114] The structure of the pixel PXij in embodiments of the present inventive concept is not limited to the structure shown in FIG. **4**. As an example, in embodiments of the present inventive concept, the pixel PXij may include the second transistor M2, the first transistor M1, and the fifth transistor M5 composed of P-type transistors, and other circuit configurations may be implemented with various types of pixel circuits.

[0115] FIG. **5** is a waveform diagram illustrating an embodiment of a method of driving the pixel of FIG. **4** during a display scan period. The display scan period may be included in an active section of a frame.

[0116] Referring to FIGS. **4** and **5**, a display scan period DSP may include a first section P**1**, a second section P**2**, a third section P**3**, and a fourth section P**4**. The first to third sections P**1** to P**3** may be set as non-emission sections, and the fourth section P**4** may be set as an emission section. A second scan signal GC, a third scan signal GI, a fourth scan signal GB, and an emission control signal EM may have voltages of the gate high voltage VGH and the second gate low voltage VGL**2**, and a first scan signal GW may have voltages of the gate high voltage VGH and the first gate low voltage VGL**1**.

[0117] During the first to third sections P**1** to P**3**, the disable emission control signal EM may be supplied to the emission control line ELk. When the disable emission control signal EM is supplied to the emission control line ELk, the sixth transistor M**6** and the seventh transistor M**7** may be turned off. When the sixth transistor M**6** and the seventh transistor M**7** are turned off, an electrical connection between the first power source line PL**1** and the light emitting element LD may be cut off, and thus, the light emitting element LD may be set to a non-light emitting state.

[0118] During the first section P**1**, the enable third scan signal GI may be supplied to the third scan line SL**3***i*. When the enable third scan signal GI is supplied to the third scan line SL**3***i*, the third transistor M**3** may be turned on. When the third transistor M**3** is turned on, the voltage of the first initialization power source Vint**1** of the third power source line PL**3** may be supplied to the first node N**1**.

[0119] During the second section P**2**, the enable second scan signal GC may be supplied to the second scan line SL**2***i*. Accordingly, the fourth transistor M**4** may be turned on. When the fourth transistor M**4** is turned on, the first transistor M**1** may be connected in the form of a diode.

[0120] In a write section P_W overlapping the second section P**2**, the enable first scan signal GW may be supplied to the first scan line SL**1***i*. When the enable first scan signal GW is supplied to the first scan line SL**1***i*, the second transistor M**2** may be turned on. When the second transistor M**2** is turned on, the data signal may be supplied from the data line DLj to the second node N**2**. Since the first transistor M**1** is connected in the form of a diode by the turned-on fourth transistor M**4**, the first node N**1** may have a voltage in which a threshold voltage of the first transistor M**1** is compensated for in the data signal.

[0121] During the third section P**3**, the enable fourth scan signal GB may be supplied to the fourth scan line SL**4***i*. When the enable fourth scan signal GB is supplied to the fourth scan line SL**4***i*, the fifth transistor M**5** and the eighth transistor M**8** may be turned on. When the fifth transistor M**5** is turned on, the voltage of the second initialization power source Vint**2** may be supplied to the first electrode of the light emitting element LD. Accordingly, the light emitting element LD may be initialized. When the eighth transistor M**8** is turned on, a voltage of the bias power source Vbias may be supplied to the second node N**2**. When the voltage of the bias power source Vbias is supplied to the second node N**2**, the first transistor M**1** may be set to an on-bias state.

[0122] In the fourth section P**4**, the enable emission control signal EM (or low-level emission control signal) may be supplied to the emission control line ELk to turn on the sixth transistor M**6** and the seventh transistor M**7**. When the sixth transistor M**6** and the seventh transistor M**7** are turned on, a current movement path flowing to the second power source line PL**2** through the first power source line PL**1**, the sixth transistor M**6**, the first transistor M**1**, the seventh transistor M**7**, and the light emitting element LD may be formed. In this case, depending on the operation of the first transistor M**1**, a driving current corresponding to the voltage of the first node N**1** may flow through the light emitting element LD, and the light emitting element LD may emit light with a luminance corresponding to the driving current.

[0123] FIG. **6** is a waveform diagram illustrating an embodiment of a method of driving the pixel of FIG. **4** during a self-scan period. A self-scan period SSP may be a section in which light is emitted while maintaining the voltage of a previously supplied data signal, and may be a period in which the image is displayed again without the frame being switched. In an embodiment, one frame may include one display scan period DSP and one or more self-scan periods SSP. One or more self-scan periods SSP may be arranged consecutively after the display scan period DSP. The self-scan period SSP may be included in a blank section of a frame.

[0124] In the self-scan period SSP, compared to the display scan period DSP, an operation of compensating a threshold voltage and an operation of writing data may be omitted, and an operation of applying a bias voltage to the first transistor M**1** (and an operation of initializing the light emitting element LD) and an operation of emitting light may be performed. The self-scan period SSP may be set to the same or similar length as the display scan period DSP. In this case, the self-scan period SSP may include a first section P**1'**, a second section P**2'**, a third section P**3'**, and a fourth section P**4'**.

[0125] Referring to FIGS. **4** and **6**, in the first section P**1'** to the third section P**3'**, the disable emission control signal EM may be supplied to the emission control line ELk. When the disable emission control signal EM is supplied to the emission control line ELk, the sixth transistor M**6** and the seventh transistor M**7** may be turned off. Accordingly, the light emitting element LD may be set to a non-light emitting state.

[0126] In an embodiment, in the first section P**1'** to the third section P**3'**, the enable first scan signal GW, the enable second scan signal GC, and the enable third scan signal GI are not supplied (alternatively, the disable scan signals GW, GC, and GI may be supplied). Accordingly, in the first section P**1'** to the third section P**3'**, the second transistor M**2**, the third transistor M**3**, and the fourth transistor M**4** may be set to a turned-off state.

[0127] In the third section P**3'**, the enable fourth scan signal GB may be supplied to the fourth scan line SL**4***i*. When the enable fourth scan signal GB is supplied to the fourth scan line SL**4***i*, the fifth transistor M**5** and the eighth transistor M**8** may be turned on.

[0128] When the fifth transistor M**5** is turned on, the voltage of the second initialization power source Vint**2** may be supplied to the first electrode of the light emitting element

LD, and thus, the light emitting element LD may be initialized. When the eighth transistor M8 is turned on, the voltage of the bias power source Vbias may be supplied to the second node N2. When the voltage of the bias power source Vbias is supplied to the second node N2, the first transistor M1 may be set to an on-bias state.

[0129] As described above, the display device **100** according to embodiments of the present inventive concept may be driven at various driving frequencies (various frame frequencies) because one frame includes the display scan period DSP and the self-scan period SSP.

[0130] FIGS. **7** and **8** are diagrams illustrating an embodiment of signals supplied during an active period and a blank period. The scan signals GW, GC, GI, and GB shown in FIGS. **7** and **8** may indicate the supply status in the display scan period DSP and the self-scan period SSP. The second scan signal GC and the third scan signal GI may be supplied from one scan driver, and thus, may be shown as one signal.

[0131] Referring to FIG. **7**, one display scan period DSP and one self-scan period SSP may be included during one frame period **1** Frame. In the display scan period DSP, the disable emission control signal EM, the enable first scan signal GW, the enable second scan signal GC, the enable third scan signal GI, and the enable fourth scan signal GB may be supplied.

[0132] In the self-scan period SS, the disable emission control signal EM and the enable fourth scan signal GB may be supplied. That is, the disable emission control signal EM and the enable fourth scan signal GB may be supplied in both the display scan period DSP and the self-scan period SSP, and the remaining scan signals GW, GC, and GI may be supplied only during the display scan period DSP.

[0133] As the driving frequency of the display device **100** decreases (for example, low-frequency driving), the number of self-scan periods SSP included in one frame period **1** Frame may increase, as shown in FIG. **8**. Accordingly, the number of times the disable emission control signal EM and the enable fourth scan signal GB are supplied during one frame period **1** Frame may also be increased.

[0134] FIG. **9** is a diagram illustrating an embodiment of a stage circuit of a first scan driver. The first scan driver **132** may include stage circuits connected to each of the first scan lines SL**11** to SL**1***n*, and the stage circuits may sequentially generate the enable first scan signal GW.

[0135] Referring to FIG. **9**, each of the stage circuits of the first scan driver **132** may include a control unit **1324** (also referred to as a control circuit) and an output unit **1322***a* (also referred to as an output circuit).

[0136] The control unit **1324** may receive a carry signal CR (or scan start signal) from a previous stage circuit and control the output unit **1322***a* in response to the carry signal CR. To this end, the control unit **1324** may additionally receive clock signals. The control unit **1324** may be implemented with various types of circuits.

[0137] The output unit **1322***a* may supply the first scan signal GW to an output terminal OUT using the gate high voltage VGH and the first gate low voltage VGL**1**. The output unit **1322***a* may supply the gate high voltage VGH or the first gate low voltage VGL**1** to the output terminal OUT in response to the control of the control unit **1324**. The output unit **1322***a* may be implemented with various types of circuits.

[0138] FIG. **10** is a diagram illustrating an embodiment of a stage circuit of the first scan driver. In describing FIG. **10**, for convenience of explanation, overlapping descriptions of the same components as those described with reference to FIG. **9** will be omitted.

[0139] Referring to FIG. **10**, the stage circuit of the first scan driver **132** may include a control unit **1324** and an output unit **1322***b*.

[0140] The output unit **1322***b* may supply the first scan signal GW to an output terminal OUT using the gate high voltage VGH and the clock signal s_CLK. A high voltage of the clock signal s_CLK may be set to the gate high voltage VGH, and a low voltage may be set to the first gate low voltage VGL**1**. The output unit **1322***b* may supply the gate high voltage VGH or the clock signal s_CLK to the output terminal OUT in response to the control of the control unit **1324**. That is, the output unit **1322***b* may output the gate high voltage VGH as a high voltage of the first scan signal GW, and output the first gate low voltage GL**1** of the clock signal s_CLK as the low voltage of the first scan signal GW. The output unit **1322***b* may be implemented with various types of circuits.

[0141] FIG. **11** is a diagram illustrating a scan driver and an emission driver according to an embodiment of the present inventive concept. A scan driver **130***a* and an emission driver **150***a* of FIG. **11** may supply a scan signal and an emission control signal to the pixel PXij shown in FIG. **12**.

[0142] Referring to FIG. **11**, the scan driver **130***a* according to an embodiment of the present inventive concept may include a first scan driver **132***a*, a second scan driver **134***a*, a third scan driver **136***a*, and a fourth scan driver **138***a*. Depending on the design, at least some of the scan drivers **132***a*, **134***a*, **136***a*, and **138***a* may be integrated into one driving circuit, module, etc.

[0143] The first scan driver **132***a* may receive a first scan start signal FLM**1** and generate an enable first scan signal while shifting the first scan start signal FLM**1** in response to a clock signal. The first scan driver **132***a* may sequentially supply the enable first scan signal to the first scan lines SL**11** to SL**1***n*. In an embodiment, the first scan driver **132***a* may supply the enable first scan signal during a display scan period of one frame.

[0144] The first scan driver **132***a* may generate a first scan signal using a first gate high voltage VGH**1** and a gate low voltage VGL. As an example, a gate-off voltage of the first scan signal supplied to an N-type transistor may be set to the gate low voltage VGL, and a gate-on voltage may be set to the first gate high voltage VGH**1**. The enable first scan signal may be set to the first gate high voltage VGH**1** so that the N-type transistor can be turned on, and a disable first scan signal may be set to the gate low voltage VGL so that the N-type transistor can be turned off.

[0145] The second scan driver **134***a* may receive a second scan start signal FLM**2** and generate an enable second scan signal while shifting the second scan start signal FLM**2** in response to a clock signal. The second scan driver **134***a* may sequentially supply the enable second scan signal to the second scan lines SL**21** to SL**2***n*. In an embodiment, the second scan driver **134***a* may supply the enable second scan signal during the display scan period and self-scan period of one frame.

[0146] The second scan driver **134***a* may generate a second scan signal using a second gate high voltage VGH**2** and the gate low voltage VGL. As an example, a gate-off voltage of the second scan signal supplied to an N-type transistor may be set to the gate low voltage VGL, and a gate-on voltage may be set to the second gate high voltage VGH**2**. The enable second scan signal may be set to the second gate high voltage VGH**2** so that the N-type transistor can be turned on, and a disable second scan signal may be set to the gate low voltage VGL so that the N-type transistor can be turned off. The first gate high voltage VGH**1** may be set to a voltage higher than the second gate high voltage VGH**2**. This will be described in further detail below.

[0147] The third scan driver **136***a* may receive a third scan start signal FLM**3** and generate an enable third scan signal while shifting the third scan start signal FLM**3** in response to a clock signal. The third scan driver **136***a* may sequentially supply the enable third scan signal to the third scan lines SL**31** to SL**3***n*. In an embodiment, the third scan driver **136***a* may supply the enable third scan signal during the display scan period of one frame.

[0148] The third scan driver **136***a* may generate a third scan signal using the first gate high voltage VGH**1** and the gate low voltage VGL. As an example, a gate-off voltage of the third scan signal supplied to an N-type transistor may be set to the gate low voltage VGL, and a gate-on voltage may be set to the first gate high voltage VGH**1**. The enable third scan signal may be set to the first gate high voltage VGH**1** so that the N-type transistor can be turned on, and a disable third scan signal may be set to the gate low voltage VGL so that the N-type transistor can be turned off.

[0149] The fourth scan driver **138***a* may receive a fourth scan start signal FLM**4** and generate an enable fourth scan signal while shifting the fourth scan start signal FLM**4** in response to a clock signal. The fourth scan driver **138***a* may sequentially supply the enable fourth scan signal to the fourth scan lines SL**41** to SLAn. In an embodiment, the fourth scan driver **138***a* may supply the enable fourth scan signal during the display scan period of one frame.

[0150] The fourth scan driver **138***a* may generate a fourth scan signal using the first gate high voltage VGH**1** and the gate low voltage VGL. As an example, a gate-off voltage of the fourth scan signal supplied to an N-type transistor may be set to the gate low voltage VGL, and a gate-on voltage may be set to the first gate high voltage VGH**1**. The enable fourth scan signal may be set to the first gate high voltage VGH**1** so that the N-type transistor can be turned on, and a disable fourth scan signal may be set to the gate low voltage VGL so that the N-type transistor can be turned off.

[0151] The emission driver **150***a* may receive an emission start signal EFLM and generate a disable emission control signal EM while shifting the emission start signal EFLM in response to a clock signal. The emission driver **150***a* may sequentially supply the disable emission control signal EM to the emission control lines EL**1** to ELo.

[0152] The emission driver **150***a* may generate an emission control signal using the first gate high voltage VGH**1** and the gate low voltage VGL. As an example, a gate-off voltage of the emission control signal supplied to an N-type transistor may be set to the gate low voltage VGL, and a gate-on voltage may be set to the first gate high voltage VGH**1**. The disable emission control signal may be set to the gate low voltage VGL so that the N-type transistor can be turned off, and an enable emission control signal may be set to the first gate high voltage VGH**1** so that the N-type transistor can be turned on.

[0153] In an embodiment, the emission driver **150***a* may supply the disable emission control signal during the display scan period and self-scan period of one frame. For example, the emission driver **150** may perform scanning once during the display scan period, and may perform scanning at least once according to the image refresh rate during the self-scan period. When the image refresh rate is reduced (that is, the frame length becomes longer), the number of repetitions of the operation in which the emission driver **150***a* supplies the disable emission control signal to each of the emission control lines EL**1** to ELo within one frame period may be increased.

[0154] FIG. **12** is a diagram illustrating a pixel according to an embodiment of the present inventive concept. FIG. **12** shows a pixel located on an i-th horizontal line and a j-th vertical line.

[0155] Referring to FIG. **12**, a pixel PXij according to an embodiment of the present inventive concept may be connected to corresponding signal lines SL**1***i*, SL**2***i*, SL**3***i*, SL**4***i*, DLj, and Elk. For example, the pixel PXij may be connected to an i-th first scan line SL**1***i*, an i-th second scan line SL**2***i*, an i-th third scan line SL**3***i*, an i-th fourth scan line SL**4***i*, a k-th emission control line Elk, and a j-th data line DLj.

[0156] The pixel PXij according to an embodiment of the present inventive concept may include a light emitting element LD and a pixel circuit that controls the amount of current supplied to the light emitting element LD.

[0157] The light emitting element LD may be connected between a first power source line PL**1***a* and a second power source line PL**2***a*. As an example, a first electrode (for example, an anode electrode) of the light emitting element LD may be connected to the first power source line PL**1***a* via a third node N**3***a*, a third transistor M**3***a* (also referred to as a first emission transistor), a second node N**2***a*, a first transistor M**1***a* (also referred to as a driving transistor), and a sixth transistor M**6***a* (also referred to as a second emission transistor), and a second electrode (for example, a cathode electrode) of the light emitting element LD may be connected to the second power source line PL**2***a*. The light emitting element LD may generate light with a luminance corresponding to the amount of current supplied from the pixel circuit.

[0158] The light emitting element LD may be, for example, an organic light emitting diode or an inorganic light emitting diode, such as, for example, a micro LED (light emitting diode) or a quantum dot light emitting diode. In addition, the light emitting element LD may be an element composed of a combination of organic and inorganic materials. FIG. **12** shows the pixel PXij including a single light emitting element LD. However, embodiments of the present inventive concept are not limited thereto. For example, in an embodiment, the pixel PXij may include a plurality of light emitting elements LD, and the plurality of light emitting elements LD may be connected in series, in parallel, or in series and parallel.

[0159] The pixel circuit may include the first transistor M**1***a* (also referred to as a driving transistor), a second transistor M**2***a*, the third transistor M**3***a* (also referred to as a first emission transistor), a fourth transistor M**4***a*, a fifth transistor M**5***a* (also referred to as a control transistor), the sixth transistor M**6***a* (also referred to as a second emission transistor), a first capacitor Csta, and a second capacitor Chold. Here, the first to sixth transistors M**1***a* to M**6***a* may be oxide semiconductor transistors. For example, an active layer (or semiconductor layer) of the first to sixth transistors M**1***a* to M**6***a* may include an oxide semiconductor layer. In an embodiment, the first to sixth transistors M1a to M6a may be N-type oxide semiconductor transistors.

[0160] A first electrode of the first transistor M1a (or driving transistor) may be connected to a second electrode of the sixth transistor M6a, and a second electrode of the first transistor M1a may be connected to the second node N2a. In addition, a first gate electrode of the first transistor M1a may be connected to a first node N1a, and a second gate electrode (or back gate electrode) of the first transistor M1a may be connected to the second node N2a. The first transistor M1a may control the amount of current supplied from a first driving power source VDD to a second driving power source VSS via the light emitting element LD in response to a voltage of the first node N1a.

[0161] The first transistor M1a may be composed of a double gate transistor including the first gate electrode and the second gate electrode. When the second gate electrode is connected to the second node N2a, a gate-source voltage and driving current of the first transistor M1a can be maintained stably.

[0162] The second transistor M2a (or write transistor) may be connected between the data line DLj and the first node N1a. In addition, a gate electrode of the second transistor M2a may be connected to a first scan line SL1i (write scan signal). The second transistor M2a may be turned on when an enable first scan signal GW (or write scan signal) is supplied to the first scan line SL1i to electrically connect the data line DLj and the first node N1a.

[0163] The third transistor M3a (or first light emitting transistor) may be connected between the second node N2a and the third node N3a (or the anode electrode of the light emitting element). The second node N2a may refer to a node where the second electrode of the first transistor M1a and a first electrode of the third transistor M3a are electrically connected to each other, and the third node N3a may refer to a node to which the first electrode of the light emitting element LD is connected. A gate electrode of the third transistor M3a may be connected to a fourth scan line SLAi. The third transistor M3a may be turned on when an enable fourth scan signal GE (or first control scan signal) is supplied to the fourth scan line SLAi.

[0164] When the third transistor M3a is turned on, the second node N2a and the third node N3a may be electrically connected to each other. Accordingly, the first transistor M1a and the light emitting element LD may be electrically connected to each other. When the third transistor M3a is turned off, the second node N2a and the third node N3a may be electrically isolated. Accordingly, a current path through which the driving current flows to the light emitting element LD may be cut off.

[0165] The fourth transistor M4a (or initialization transistor) may be connected between the third node N3a and a fourth power source line PL4a (or initialization power source line). In addition, a gate electrode of the fourth transistor M4a may be connected to a second scan line SL2i (or initialization scan signal). The fourth transistor M4a may be turned on when an enable second scan signal GI (or initialization scan signal) is supplied to the second scan line SL2i to electrically connect the fourth power source line PL4a and the third node N3a.

[0166] When the fourth power source line PL4a and the third node N3a are electrically connected to each other, a voltage of the initialization power source VINT from the fourth power source line PL4a may be supplied to the third node N3a. Then, a parasitic capacitor equivalently formed in the light emitting element LD may be discharged, and thus, the black expression ability can be improved.

[0167] The fifth transistor M5a may be connected between a third power source line PL3a (or reference power source line) and the first node N1a. In addition, a gate electrode of the fifth transistor M5a may be connected to a third scan line SL3i. The fifth transistor M5a may be turned on when an enable third scan signal GR (or second control scan signal) is supplied to the third scan line SL3i to electrically connect the third power source line PL3a and the first node N1a. When the third power source line PL3a and the first node N1a are electrically connected to each other, a voltage of the reference power source VREF may be supplied to the first node N1a.

[0168] The sixth transistor M6a (or second light emitting transistor) may be connected between the first power source line PL1a and the first electrode of the first transistor M1a. In addition, a gate electrode of the sixth transistor M6a may be connected to the emission control line ELk. The sixth transistor M6a may be turned off when a disable emission control signal EM (or low level emission control signal EM) is supplied to the emission control line ELk, and may be turned on in other cases. When the sixth transistor M6a is turned on, a current path through which the driving current flows through the pixel PXij may be formed.

[0169] The first capacitor Csta may be connected between the first node N1a and the second node N2a. A voltage corresponding to the data signal may be stored in the first capacitor Cst.

[0170] The second capacitor Chold may be connected between the first power source line PL1a and the second node N2a. The second capacitor Chold may stabilize a voltage of the second node N2a.

[0171] A gate-on voltage (or high voltage) of the first scan signal GW must be set to a voltage higher than the data signal supplied to the data line DLj. As an example, when a voltage of the data signal is set to about 1.5V to about 7V, the gate-on voltage of the first scan signal GW may be set to about 10V in consideration of a threshold voltage of the second transistor M2a. Here, the gate-on voltage of the first scan signal GW may be set to the first gate high voltage VGH1.

[0172] The first gate high voltage VGH1 may be set to the gate-on voltage of the first scan signal GW, and gate-on voltages of the remaining transistors M3a, M5a, and M6a may also be set to the first gate high voltage VGH1.

[0173] In an embodiment of the present inventive concept, a gate-on voltage of a second scan signal GI may be set to the second gate high voltage VGH2, which is lower than the first gate high voltage VGH1. The fourth transistor M4a may receive a voltage of the initialization power source VINT that is lower than the data signal. Accordingly, the gate-on voltage of the second scan signal GI may be set to the second gate high voltage VGH2 that is lower than the first gate high voltage VGH1. In this case, the voltage swing width of the second scan signal GI may be reduced, and thus, power consumption can be reduced.

[0174] In an embodiment of the present inventive concept, a gate-on voltage of a third scan signal GR may be set to the second gate high voltage VGH2. As an example, when the voltage of the reference power source VREF is set to a voltage similar to the lowest grayscale of the data signal, the gate-on voltage of the third scan signal GR may be set to the second gate high voltage VGH2.

[0175] The structure of the pixel PXij is not limited to the structure shown in FIG. **14**. For example, in an embodiment of the present inventive concept, the pixel PXij may include the second transistor M**2***a*, the first transistor M**1***a*, and the fourth transistor M**4***a* composed of N-type transistors, and other circuit configurations may be implemented with various types of pixel circuits.

[0176] FIG. **13** is a diagram illustrating an embodiment of a method of driving the pixel of FIG. **12** during a display scan period. The display scan period DSP may be included in an active section of a frame.

[0177] Referring to FIGS. **12** and **13**, a method of driving the pixel PXij may include a first period Pla, a second period P**2***a*, a third period P**3***a*, a fourth period P**4***a*, and a fifth period P**5***a*.

[0178] The first scan signal GW, the third scan signal GR, a fourth scan signal GE, and an emission control signal EM may have voltages of the first gate high voltage VGH**1** and the gate low voltage VGL, and the second scan signal GI may have voltages of the second gate high voltage VGH**2** and the gate low voltage VGL.

[0179] The first period Pla may be a period in which the first capacitor Csta is initialized. The second period P**2***a* may be a period in which a threshold voltage of the first transistor M**1***a* is compensated for. The third period P**3***a* may be a period in which the voltage of the data signal is stored in the pixel PXij. The fourth period P**4***a* may be a period in which the light emitting element LD is initialized. The fifth period P**5***a* may be a period in which the pixel PXij (or light emitting element LD) emits light.

[0180] During the first period Pla, the enable second scan signal GI may be supplied to the second scan line SL**2***i*, the enable third scan signal GR may be supplied to the third scan line SL**3***i*, and the enable fourth scan signal GE may be supplied to the fourth scan line SL**4***i*. In addition, the disable emission control signal EM may be supplied to the emission control line ELk.

[0181] When the disable emission control signal EM is supplied to the emission control line ELk, the sixth transistor M**6***a* may be turned off. When the sixth transistor M**6***a* is turned off, the electrical connection between the first power source line PL**1***a* and the first transistor M**1***a* may be cut off, and thus, the light emitting element LD may be set to a non-light emitting state.

[0182] When the enable second scan signal GI is supplied to the second scan line SL**2***i*, the fourth transistor M**4***a* may be turned on. When the fourth transistor M**4***a* is turned on, the voltage of the initialization power source VINT may be supplied to the third node N**3***a*. When the enable fourth scan signal GE is supplied to the fourth scan line SL**4***i*, the third transistor M**3***a* may be turned on. When the third transistor M**3***a* is turned on, the voltage of the initialization power source VINT of the third node N**3***a* may be supplied to the second node N**2***a*.

[0183] When the enable third scan signal GR is supplied to the third scan line SL**3***i*, the fifth transistor M**5***a* may be turned on. When the fifth transistor M**5***a* is turned on, the voltage of the reference power source VREF may be supplied to the first node N**1***a*. When the voltage of the reference power source VREF is supplied to the first node N**1***a* and the voltage of the initialization power source VINT is supplied to the second node N**2***a*, the first capacitor Csta and the second capacitor Chold may be initialized. That is, the first period Pla may be a period in which the pixel PXij is initialized so that it is not affected by the data signal supplied in a previous frame period.

[0184] During the second period P**2***a*, the supply of the disable emission control signal EM to the emission control line ELk may be stopped (or the enable (high level) emission control signal EM may be supplied to the emission control line ELk), and the enable third scan signal GR may be supplied to the third scan line SL**3***i*. The enable third scan signal GR supplied to the third scan line SL**3***i* may be supplied during the first period Pla and the second period P**2***a*.

[0185] When the supply of the disable emission control signal EM to the emission control line ELk is stopped, the sixth transistor M**6***a* may be turned on. Accordingly, the voltage of the first driving power source VDD may be supplied to the first electrode of the first transistor M**1***a*. When the enable third scan signal GR is supplied to the third scan line SL**3***i*, the fifth transistor M**5***a* may be turned on. Accordingly, the voltage of the reference power source VREF may be supplied to the first node N**1***a*.

[0186] Here, the voltage of the reference power source VREF may be set so that the first transistor M**1***a* can be turned on. Accordingly, the voltage of the second node N**2***a* may be increased in response to the current supplied from the first transistor M**1***a*. The voltage of the second node N**2***a* may be increased to a value obtained by subtracting an absolute value of the threshold voltage of the first transistor M**1***a*. That is, during the second period P**2***a*, a voltage corresponding to the threshold voltage of the first transistor M**1***a* may be stored in the first capacitor Csta.

[0187] The width of the second period P**2***a* may be determined by the supply time of the enable emission control signal EM and the enable third scan signal GR. That is, in an embodiment of the present inventive concept, the compensation time (that is, the second period P**2***a*) of the threshold voltage of the first transistor M**1***a* may be controlled using the supply time of the enable emission control signal EM and the enable third scan signal GR.

[0188] During the third period P**3***a*, the disable emission control signal EM may be supplied, and thus, the sixth transistor M**6***a* may be turned off. During the third period P**3***a*, the enable first scan signal GW may be supplied to the first scan line SL**1***i*. When the enable first scan signal GW is supplied to the first scan line SL**1***i*, the second transistor M**2***a* may be turned on. When the second transistor M**2***a* is turned on, the data signal from the data line DLj may be supplied to the first node N**1***a*.

[0189] Voltages of the first node N**1***a* and the second node N**2***a* during the third period P**3***a* may be expressed as Equation 2 below.

$$VN1a = Vdata \qquad \text{[Equation 2]}$$

$$VN2a = VREF - Vth1$$

[0190] In Equation 2, Vdata may refer to the voltage of the data signal, and Vth**1** may refer to the threshold voltage of the first transistor M**1***a*.

[0191] In Equation 2, for convenience of description, it is described that the second node N**2***a* maintains a voltage of

VREF-Vth**1** during the third period P**3**a, but embodiments of the present inventive concept are not limited thereto.

[0192] As an example, during the third period P**3**a, the voltage of the first node N**1**a may be changed from the voltage of the reference power source VREF to the voltage of the data signal Vdata, and the voltage of the second node N**2**a may also be changed by coupling of the first capacitor Csta. However, the voltage of the second node N**2**a may be changed in response to a ratio of the first capacitor Csta and the second capacitor Chold. Accordingly, the amount of change in voltage at the second node N**2**a can be reduced. Hereafter, for convenience of description, it is assumed that the second node N**2**a maintains a voltage of VREF-Vth**1** during the third period P**3**a.

[0193] During the fourth period P**4**a, the supply of the disable emission control signal EM may be maintained, and the enable second scan signal GI may be supplied to the second scan line SL**2**i. When the enable second scan signal GI is supplied to the second scan line SL**2**i, the fourth transistor M**4**a may be turned on. When the fourth transistor M**4**a is turned on, the voltage of the initialization power source VINT may be supplied to the third node N**3**a. When the voltage of the initialization power source VINT is supplied to the third node N**3**a, the first electrode of the light emitting element LD (or the parasitic capacitor of the light emitting element LD) may be initialized with the voltage of the initialization power source VINT.

[0194] During the fifth period P**5**a, the enable emission control signal EM may be supplied to the emission control line ELk. When the enable emission control signal EM is supplied to the emission control line ELk, the sixth transistor M**6**a may be turned on. When the sixth transistor M**6**a is turned on, the first power source line PL**1** and the first transistor M**1**a may be electrically connected to each other.

[0195] Also, during the fifth period P**5**a, the enable fourth scan signal GE may be supplied to the fourth scan line SL**4**i. When the enable fourth scan signal GE is supplied to the fourth scan line SL**4**i, the third transistor M**3**a may be turned on. When the third transistor M**3**a is turned on, the second node N**2**a and the third node N**3**a may be electrically connected to each other.

[0196] In this case, the first transistor M**1**a may supply a driving current corresponding to the voltage of the first node N**1**a from the first driving power source VDD to the second driving power source VSS via the light emitting element LD. Then, during the fifth period P**5**a, the light emitting element LD may generate light with a luminance corresponding to the driving current. The fifth period P**5**a may be an emission period.

[0197] FIG. **14** is a diagram illustrating an embodiment of a method of driving the pixel of FIG. **12** during a self-scan period.

[0198] Referring to FIGS. **12** to **14**, in an embodiment of the present inventive concept, a frame period may include one display scan period DSP and at least one self-scan period SSP.

[0199] The display scan period DSP may be a period in which the voltage of the data signal is stored in the pixels PX, and the driving signals of FIG. **13** described above may be supplied. That is, the display scan period DSP may include the first period Pla, the second period P**2**a, the third period P**3**a, the fourth period P**4**a, and the fifth period P**5**a shown in FIG. **13**. The enable scan signals GW, GI, GR, and GE and the disable emission control signal EM may be supplied in each corresponding period (at least one period among Pla to P**5**a).

[0200] The self-scan period SSP may be a period in which the pixels PX are set to a non-light emitting state for a part of the period while maintaining the data signal supplied in the display scan period DSP. One or more of self-scan periods SSP may be included in the frame period. When the self-scan period SSP is included in one frame period, the pixels PX may be set to a non-light emitting state at regular intervals. Accordingly, the quality of a moving image can be improved.

[0201] As shown in FIG. **14**, the self-scan period SSP may include a first period Pla', a second period P**2**a', a third period P**3**a', a fourth period P**4**a', and a fifth period P**5**a' corresponding to the first period Pla, the second period P**2**a, the third period P**3**a, the fourth period P**4**a, and the fifth period P**5**a of FIG. **13**.

[0202] During the first period P**1**a' to the fourth period P**4**a' of the self-scan period SSP, the disable emission control signal EM may be supplied to the emission control line ELk. When the disable emission control signal EM is supplied to the emission control line ELk, the sixth transistor M**6**a may be turned off. Accordingly, during the first period P**1**a' to the fourth period P**4**a', the light emitting element LD may be set to a non-light emitting state.

[0203] During the first period P**1**a' and the fourth period P**4**a' of the self-scan period SSP, the enable second scan signal GI may be supplied to the second scan line SL**2**i. When the enable second scan signal GI is supplied to the second scan line SL**2**i, the fourth transistor M**4**a may be turned on. Accordingly, the first electrode of the light emitting element LD may be initialized with the voltage of the initialization power source VINT.

[0204] During the fifth period P**5**a' of the self-scan period SSP, the enable emission control signal EM may be supplied to the emission control line ELk. When the enable emission control signal EM is supplied to the emission control line ELk, the sixth transistor M**6**a may be turned on. When the sixth transistor M**6**a is turned on, the first power source line PL**1**a and the first transistor M**1**a may be electrically connected to each other.

[0205] In this case, the first transistor M**1**a may supply a driving current corresponding to the voltage of the first node N**1**a from the first driving power source VDD to the second driving power source VSS via the light emitting element LD. Then, during the fifth period P**5**a', the light emitting element LD may generate light with a luminance corresponding to the driving current.

[0206] FIG. **15** is a diagram illustrating an embodiment of a stage circuit of a second scan driver shown in FIG. **11**. The second scan driver **134**a may include stage circuits connected to each of the second scan lines SL**21** to SL**2**n. The stage circuits may sequentially generate the enable second scan signal GI.

[0207] Referring to FIG. **15**, each of the stage circuits of the second scan driver **134**a may include a control unit **184** and an output unit **182**.

[0208] The control unit **184** may receive a carry signal CR (or scan start signal) from a previous stage circuit and control the output unit **182** in response to the carry signal CR. To this end, the control unit **184** may additionally receive clock signals. The control unit **184** may be implemented with various types of circuits.

[0209] The output unit **182** may supply the second scan signal GI to an output terminal OUT using the second gate high voltage VGH**2** and the gate low voltage VGL. The output unit **182** may supply the second gate high voltage VGH**2** or the gate low voltage VGL to the output terminal OUT in response to the control of the control unit **184**. The output unit **182** may be implemented with various types of circuits. Additionally, the second gate high voltage VGH**2** and/or the gate low voltage VGL may be supplied to the output unit **182** in the form of a pulse signal.

[0210] A display device according to embodiments of the present inventive concept may reduce power consumption by reducing a gate high voltage of a scan signal or increasing a gate low voltage of the scan signal.

[0211] However, effects of the present inventive concept are not limited to the above-described effects, and may be variously extended without departing from the spirit and scope of the present inventive concept.

[0212] While the present inventive concept has been particularly shown and described with reference to embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and detail may be made therein without departing from the spirit and scope of the present inventive concept as defined by the following claims.

What is claimed is:

1. A display device, comprising:

a pixel unit including a plurality of pixels,

wherein each of the pixels includes:

a light emitting element;

a driving transistor configured to control an amount of current flowing from a first power source line to a second power source line via the light emitting element in response to a voltage of a first node;

a write transistor connected between a first electrode of the driving transistor and a data line, and configured to be turned on or off in response to a write scan signal; and

an initialization transistor connected between an anode electrode of the light emitting element and an initialization power source line, and configured to be turned on or off in response to an initialization scan signal,

wherein a low voltage of the write scan signal is set to a first gate low voltage, and a low voltage of the initialization scan signal is set to a second gate low voltage that is different from the first gate low voltage.

2. The display device of claim **1**, wherein the write transistor and the initialization transistor are P-type transistors.

3. The display device of claim **1**, wherein a high voltage of the write scan signal and a high voltage of the initialization scan signal are set to a gate high voltage.

4. The display device of claim **1**, wherein the first gate low voltage is a voltage higher than the second gate low voltage.

5. The display device of claim **1**, further comprising:

a data driver configured to supply a data signal to the data line;

a first scan driver configured to supply the write scan signal to a write scan line; and

a second scan driver configured to supply the initialization scan signal to an initialization scan line.

6. The display device of claim **5**, wherein the first scan driver includes a plurality of stage circuits, and

wherein each of the stage circuits includes:

an output unit configured to generate the write scan signal using a gate high voltage and the first gate low voltage; and

a control unit configured to control the output unit.

7. The display device of claim **5**, wherein the first scan driver includes a plurality of stage circuits, and

wherein each of the stage circuits includes:

an output unit configured to generate the write scan signal using a gate high voltage and a clock signal; and

a control unit configured to control the output unit.

8. The display device of claim **7**, wherein a high voltage of the clock signal is set to the gate high voltage, and a low voltage of the clock signal is set to the first gate low voltage.

9. A display device, comprising:

a pixel unit including a plurality of pixels,

wherein each of the pixels includes:

a light emitting element;

a driving transistor configured to control an amount of current flowing from a first power source line to a second power source line via the light emitting element in response to a voltage of a first node;

a write transistor connected between the first node and a data line, and configured to be turned on or off in response to a write scan signal; and

an initialization transistor connected between an anode electrode of the light emitting element and an initialization power source line, and configured to be turned on or off in response to an initialization scan signal,

wherein a high voltage of the write scan signal is set to a first gate high voltage, and a high voltage of the initialization scan signal is set to a second gate high voltage.

10. The display device of claim **9**, wherein the write transistor and the initialization transistor are N-type transistors.

11. The display device of claim **9**, wherein a low voltage of the write scan signal and a low voltage of the initialization scan signal are set to a gate low voltage.

12. The display device of claim **9**, wherein the first gate high voltage is a voltage higher than the second gate high voltage.

13. The display device of claim **9**, wherein each of the pixels further includes:

a first emission transistor connected between a second electrode of the driving transistor and the anode electrode of the light emitting element, and configured to be turned on or off by a first control scan signal;

a second emission transistor connected between the first power source line and a first electrode of the driving transistor, and configured to be turned on or off by an emission control signal;

a control transistor connected between a reference power source line and the first node, and configured to be turned on or off by a second control scan signal;

a first capacitor connected between the first node and the second electrode of the driving transistor; and

a second capacitor connected between the first power source line and the second electrode of the driving transistor, wherein the first emission transistor, the second emission transistor, and the control transistor are N-type transistors.

14. The display device of claim **13**, wherein a high voltage of each of the first control scan signal, the second control scan signal, and the emission control signal is set to the first gate high voltage, and a low voltage of each of the first control scan signal, the second control scan signal, and the emission control signal is set to a gate low voltage.

15. The display device of claim **13**, wherein a high voltage of each of the first control scan signal and the emission control signal is set to the first gate high voltage, and a low voltage of each of the first control scan signal and the emission control signal is set to a gate low voltage, and

wherein a high voltage of the second control scan signal is set to the second gate high voltage, and a low voltage of the second control scan signal is set to the gate low voltage.

16. The display device of claim **9**, further comprising:

a data driver configured to supply a data signal to the data line;

a first scan driver configured to supply the write scan signal to a write scan line; and

a second scan driver configured to supply the initialization scan signal to an initialization scan line.

17. The display device of claim **16**, wherein the second scan driver includes a plurality of stage circuits, and

wherein each of the stage circuits includes:

an output unit configured to generate the initialization scan signal using the second gate high voltage and a gate low voltage; and

a control unit configured to control the output unit.

18. A display device, comprising:

a pixel unit including a plurality of pixels connected to a plurality of first scan lines, a plurality of second scan lines, and a plurality of data lines;

a power source generator configured to generate a gate high voltage, a first gate low voltage, and a second gate low voltage using a first power source input from outside of the power source generator and a second power source having a voltage lower than the first power source;

a scan driver configured to generate a first scan signal to be supplied to the first scan lines using the gate high voltage and the first gate low voltage, and configured to generate a second scan signal to be supplied to the second scan lines using the gate high voltage and the second gate low voltage; and

a data driver configured to supply a data signal to the data lines.

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